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Markov-switching Ornstein-Uhlenbeck process for modelling crude oil time spreads

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The work contained in this thesis is my own work unless otherwise stated.

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Contents

1.	Introduction	2
2.	Methods	4
2.1.	Classic Ornstein-Uhlenbeck	4
2.2.	Heavy tails, Student's t residuals	5
2.3.	Markov Regime Switching OU (with Student's t)	9
2.4.	Evaluation Metrics	14
2.5.	Programming the model	15
3.	Model design choices	15
3.1.	Model architecture	15
3.2.	Parameter constraints and initialization	16
3.3.	Optimization mechanics	17
3.4.	HMM EM psuedo code	18
3.5.	Backtesting/execution	19
4.	Results	21
4.1.	In-sample diagnostics	21
4.2.	Out-of-sample results	24
4.2.1.	Student's t vs Gaussian OU model results	24
4.2.2.	Baseline Student's t OU	27
4.2.3.	Regime Switching HMM-OU results	28
5.	Discussion	34
6.	Endmatter	34
A.	Supplementary Materials	Supplementary Ma
B.	Definitions, theorems and examples	Supplementary Ma

Abstract

A regime-switching hidden Markov Ornstein-Uhlenbeck (OU) framework was used in an attempt to model Brent crude oil time spreads. As a baseline, a single-regime OU with Student's t residuals is estimated, allowing the model to accommodate heavy tails and extreme observations inherent to oil markets better than a conventional Gaussian OU specification. The framework is then extended to a three-regime Hidden Markov Model to overcome the inadequacy of using a single regime to describe distinct and fundamentally different sets of market conditions. Each regime is characterized by its own set of OU parameters and they are estimated via Expectation-Maximization algorithm due to the regimes being latent and the observations cannot be directly assigned to individual OU processes. Forecasts are then constructed from the probability-weighted mixture of regime-specific OU distributions. Model performance is assessed through an out-of-sample rolling backtest and compared across the models to determine if explicitly catering for different regimes provide incremental forecasting value and uncertainty quantification over a conventional mean-reverting single regime OU model and other naive forecast benchmarks.

1. Introduction

This paper attempts to extend and modify the classical Ornstein-Uhlenbeck (OU) process to more realistically model real-world data, specifically in the Brent crude oil market. The OU process was originally introduced by [Ornstein and Uhlenbeck \(1930\)](#) and has been subsequently adopted widely in financial modelling ([Vasicek \(1977\)](#)) and commodity markets, where mean reversion forces exist due to the influence of physical supply, demand and inventory adjustment mechanisms. [Schwartz \(1997\)](#) demonstrated that commodity prices exhibit stochastic mean-reverting behaviour, while [Gibson and Schwartz \(1990\)](#) incorporated convenience yields into commodity valuation models. The core property of an OU process is mean reversion and is used to describe systems that fluctuate randomly but are constantly being pulled back towards a long-term equilibrium level. It assumes that deviations from equilibrium are temporary and decay exponentially toward the long-run mean.

However, a simple single-regime OU model with Gaussian assumptions prove to be inadequate in modelling oil markets where volatility, equilibrium levels and adjustment speeds vary over time and also exhibit heavy tails and experiences crisis and shocks. Crude oil spreads are also governed by physical storage economics, transport bottlenecks and inventory cycles. As such, a single-regime OU process would be misspecified as it cannot simultaneously model a calm center and violent extremes. This motivates the use of Student's t residuals and regime-switching extensions to a baseline OU to improve forecast and uncertainty quantification.

The foundational regime-switching paradigm was first established in macroeconomic business cycles by [Hamilton \(1989\)](#) and subsequently adapted to energy term structures by [Almansour \(2016\)](#) and [Yuan and Kwon \(2025\)](#). [Fong and See \(2002\)](#) provides evidence that commodity futures volatility is better represented by models allowing multiple regimes. Similarly, [Hamilton and Susmel \(1994\)](#) introduces regime-switching volatility dynamics and demonstrates that volatility clusters differently across latent regimes. As established by [Perron \(1989\)](#), a mixture of stationary processes with different means mathematically mimics a unit root (random walk).

This study contributes to the existing literature on commodity price modelling by developing a regime-switching OU framework with heavy tails to model Brent crude oil time spreads specifically. While there is extensive literature establishing the prevalence of mean reversion, heavy tails and regime switching in commodity markets individually, an empirical application applying a combination of the above to forecast and model the behavior of crude oil time spreads specifically has not been well explored. Furthermore, the study is supported by a back-testing framework to ensure robustness of theory applied empirically. A rolling out-of-sample forecasting framework is developed to determine if a mean-reverting stochastic process is able to outperform a naive random walk benchmark, and if a regime-switching model is able to outperform a single-regime mean reverting model.

2. Methods

2.1. Classic Ornstein-Uhlenbeck

$$dx_t = \theta(\mu - x_t)dt + \sigma dW_t \quad (1)$$

The classic continuous-time Ornstein–Uhlenbeck process first introduced by [Ornstein and Uhlenbeck \(1930\)](#) is defined by the stochastic differential equation (SDE) in Equation (1), where

x_t is the value of the process at time t

μ is the long term mean which the process reverts to

θ is the speed of mean reversion

σ is the volatility

dW_t is a Wiener process (brownian motion) where dW_t is distributed as $\mathcal{N}(0, dt)$

In the context of modelling crude oil time spreads, x_t represents the value (price) of the time spread at time t , μ represents the long-term equilibrium level of the time spread, θ represents the speed at which the time spread reverts back to the mean, σ represents the volatility of the time spread's daily fluctuations and dW_t represents the random and unpredictable component of daily market shocks. Using historical brent crude oil time spread data, the parameters θ , μ , and σ are estimated using Maximum Likelihood Estimation (MLE)

The solution to the SDE in equation (1) is

$$x_{t+h} = \mu + (x_t - \mu)e^{-\theta h} + \int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u \quad (2)$$

where the future value is determined by:

1) the long-term mean, plus 2) the current deviation from the mean ($(x_t - \mu)$) decayed exponentially over horizon h , plus 3) the accumulated sum of random shocks. $e^{-\theta(t+h-u)}$ is the decay weight applied to past shocks. Since the process is assumed to be mean-reverting, this factor decays the impact of old shocks.

When forecasting h steps ahead from the current observation x_t , the conditional expectation of the exact solution in equation 2 is taken. As the stochastic Ito integral $\int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u$ has a conditional expected value of 0, the future expected value depends entirely only on the first deterministic drift component:

$$E[x_{t+h}|x_t] = \mu + (x_t - \mu)e^{-\theta h} \quad (3)$$

As the forecast horizon h grows, the term $e^{-\theta h}$ decays, pulling the expected value back to the long-term equilibrium mean. As $h \rightarrow \infty$, the expectation converges precisely to the long-term mean. Conversely, for short horizons ($h \rightarrow 0$), the forecast starts near the current observed spread x_t .

The OU conditional variance is given by:

$$Var[x_{t+h}|x_t] = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta h}) \quad (4)$$

In the short term, uncertainty grows quickly, driven by the random shocks σ^2 . As h increases, the stationary variance is $\frac{\sigma^2}{2\theta}$, which determines the maximum width of the risk and uncertainty around the forecast. If θ is high, the variance bound is tighter as mean reversion is strong. This is in contrast to a random walk process where variance is unbounded, as when $\theta = 0$ in equation 4, the variance is undefined as the process is non stationary whereas in an OU, the force of mean reversion eventually counteracts the random shocks. Hence the OU confidence intervals widens with the forecast horizon before asymptotically reaching the unconditional OU variance. When $\theta = 0$, the OU drift term in equation 1, $\theta(\mu - x_t)dt = 0$ and the process simplifies to a random walk and the conditional mean in equation 3 becomes $E[x_{t+h}|x_t] = x_t$, as there is no mean reversion and the best guess for any future values x_{t+h} is simply the value which the process started with.

The half-life, $\frac{\ln(2)}{\theta}$ represents the time it takes, on average, for the OU process to decay halfway back to its long-term mean from the current deviation. A short half-life indicates rapid correction while a long half-life represents persistent deviations.

2.2. Heavy tails, Student's t residuals

Although the baseline OU model is suited for capturing the mean-reverting behavior of oil time spreads, its Gaussian assumption is inadequate in terms of quantifying the forecast errors around extreme market events. Empirically, oil time spreads occasionally experiences large market shocks and such tail risks are underestimated under a Gaussian OU and results in underestimating the probability of large forecast errors and produce forecast intervals that are too narrow during periods of crisis. To address this limitation, the Gaussian errors in the OU model is replaced by a Student's t innovation. This preserves the mean-reverting structure of the OU model while allowing for heavier-tailed forecast errors.

Gaussian OU transition:

Under the standard Gaussian OU model, the exact discrete-time transition over an interval h , implied by Equation 1, is:

$$x_{t+h} = \mu + (x_t - \mu)e^{-\theta h} + s \cdot Z \quad (5)$$

where Z is a standard Normal random variable: $Z \sim \mathcal{N}(0, 1)$ and s is the standard deviation of the OU process:

$$s = \sigma \sqrt{\frac{1 - e^{-2\theta h}}{2\theta}} \quad (6)$$

with conditional density:

$$f_{Normal}(x_{t+h}|x_t) = \frac{1}{s\sqrt{2\pi}} \exp\left(-\frac{(x_{t+h} - m)^2}{2s^2}\right) \quad (7)$$

(where $m = \mu + (x_t - \mu)e^{-\theta h}$ is the expected mean) and the OU variance is:

$$Var(x_{t+h}|x_t) = s^2 = \frac{\sigma^2}{2\theta}(1 - e^{-2\theta h}) \quad (8)$$

Student's t transition:

The Gaussian assumption in (5) is replaced with by a standard Student's t random variable and the corresponding transition equation becomes:

$$x_{t+h} = \mu + (x_t - \mu)e^{-\theta h} + s \cdot T_\nu \quad (9)$$

Where: T_ν is a standard Student's t random variable with ν degrees of freedom: $T_\nu \sim t(\nu)$

The corresponding probability density of observing x_{t+h} given x_t is then:

$$f_{Student's t}(x_{t+h}|x_t) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) s\sqrt{\pi\nu}} \left(1 + \frac{1}{\nu} \left(\frac{x_{t+h} - m}{s}\right)^2\right)^{-\frac{\nu+1}{2}} \quad (10)$$

Where $\Gamma(z) = \int_0^\infty e^{-x} x^{z-1} dx$ is the standard Gamma function and ν is the degrees of freedom.

The Student's t OU Variance is:

$$Var(x_{t+h}|x_t) = s^2 \left(\frac{\nu}{\nu-2}\right) = \left[\frac{\sigma^2}{2\theta}(1 - e^{-2\theta h})\right] \left(\frac{\nu}{\nu-2}\right), \nu > 2 \quad (11)$$

As $\frac{\nu}{\nu-2}$ is always strictly greater than 1, the Student's t process structurally assumes a higher variance than a standard OU process with the same σ and θ parameters. As the variance is higher, the 95% confidence intervals are also wider, but more realistic than a standard OU during times of stress.

Maximum Likelihood Estimation:

The Log-Likelihood function is:

$$\mathcal{LL} = \sum_{t=1}^{N-1} \ln f(x_{t+h} | x_t)$$

$$\mathcal{LL} = (N-1) \left[\ln \Gamma\left(\frac{\nu+1}{2}\right) - \ln \Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2} \ln(\pi\nu) - \ln s \right] - \frac{\nu+1}{2} \sum_{t=1}^{N-1} \ln \left[1 + \frac{(x_{t+1} - m_{t+h})^2}{\nu s^2} \right] \quad (12)$$

The parameter vector is estimated by maximum likelihood:

$$\left(\hat{\theta}, \hat{\mu}, \hat{s}, \hat{\nu}\right) = \arg \max_{\theta, \mu, s, \nu} \ell(\theta, \mu, s, \nu)$$

Difference in forecast errors and robustness to outliers:

The difference between the Gaussian and Student's t specifications becomes distinct for large forecast errors. For the Student's t in equation (13), the penalty term for the errors grow logarithmically rather than quadratically in the Gaussian case in equation (14) due to the difference in log-likelihoods:

$$\log \mathcal{L} \propto -\frac{\nu+1}{2} \sum_{i=1}^n \log \left(1 + \frac{e_i^2}{\nu s^2} \right) \quad (13)$$

$$\log \mathcal{L} = \sum_{i=1}^n \left[-\frac{1}{2} \log(2\pi\sigma^2) - \frac{e_i^2}{2\sigma^2} \right] \quad (14)$$

Using the chain rule on equation 13, the derivative of the Student's t penalty with respect to the error is proportional to:

$$\frac{\partial}{\partial e_i} \log \left(1 + \frac{e_i^2}{\nu s^2} \right) \propto \frac{e_i}{1 + \frac{e_i^2}{\nu s^2}} \quad (15)$$

For small errors, $e_i \approx 0 \implies \frac{e_i}{1 + \frac{e_i^2}{\nu s^2}} \approx e_i$ as denominator approaches 1 and the gradient simplifies to e_i , implying that for quiet, normal market days around the centre of the distribution, the Student's t model behaves similarly to the Gaussian. For large errors $|e_i| \rightarrow \infty$, $\frac{e_i}{1 + \frac{e_i^2}{\nu s^2}} \rightarrow 0$ as

the squared term in the denominator grows much faster than the linear term in the numerator, causing the gradient to approach 0. This implies that extreme observations from market shocks receive less incremental influence on the likelihood gradient as compared to a Gaussian squared error objective, making the Student's t specification more robust to occasional market shocks and preventing extreme outliers from corrupting structural estimates. In the case of a Gaussian, a sufficiently large outlier has a disproportionate influence on parameter estimation as the penalty imposed on those outlier increases quadratically with the size of the error and over-dominate other other points and forces the optimizer to drag the long-term mean higher in order to avoid massive squared penalties on isolated shocks.

Difference in tail behavior:

The probability of extreme observations decline very rapidly in the Gaussian specification as the density (in equation 7) contains the exponential term $\exp\left(-\frac{e^2}{2s^2}\right)$, whereas in the Student's t density in equation 12, it declines polynomially as it has the form $\left(1 + \frac{e^2}{\nu s^2}\right)^{-\frac{\nu+1}{2}}$. This means that the Student's t assigns substantially more probability to extreme observations than the Gaussian.

Confidence intervals:

The Student's t distribution has variance $s^2 \frac{\nu}{\nu-2}$, $\nu > 2$. By contrast, $\text{Var}(Z) = 1$ for a standard normal random variable.

Therefore, if s is the Student's t scale parameter,

$$\text{Var}(x_{t+h} | x_t) = s^2 \frac{\nu}{\nu-2} \quad (16)$$

For the Gaussian OU model,

$$\text{Var}(x_{t+h} | x_t) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta h}) \quad (17)$$

The Student's t model therefore allows a larger probability mass in the tails and produces wider prediction intervals. For a two-sided $100(1 - \alpha)\%$ prediction interval, the Student's t interval is

$$m_{t+h} \pm t_{\nu, 1-\alpha/2} \sqrt{\text{Var}(x_{t+h} | x_t)} \quad (18)$$

Hence, a 95% interval uses $m_{t+h} \pm t_{\nu, 0.975} \sqrt{\text{Var}(x_{t+h} | x_t)}$. For small ν , the critical value $t_{\nu, 0.975}$ is substantially larger than the Gaussian value 1.96. For example, $t_{3, 0.975} \approx 3.18$.

Consequently, the Student's t model produces wider prediction intervals when the estimated degrees of freedom are low, reflecting greater uncertainty associated with extreme movements. This means that the forecast interval widens and prevents the model from triggering premature or false trading signals during stress periods as easily as larger price swings are being accounted for.

Impact on point forecast:

Notably, replacing Gaussian errors with Student's t errors does not change the functional form of the conditional OU mean: $\hat{x}_{t+h} = \hat{\mu} + (x_t - \hat{\mu})e^{-\hat{\theta}h}$. However, the estimated parameters could differ because they are obtained by maximizing different likelihood functions.

Under the Gaussian OU model,

$$(\theta^G, \mu^G, \sigma^G) = \arg \max \ell_{\text{Normal}}$$

Under the Student's t OU model,

$$(\theta^T, \mu^T, s^T, \nu) = \arg \max \ell_{\text{Student's t}}$$

Therefore, although the forecasting equation has the same mean-reverting structure, the resulting point forecast can differ because

$$(\theta^G, \mu^G) \neq (\theta^T, \mu^T)$$

Hence, the main benefit of the Student's t extension is not that of a different forecasting centre, but a more realistic representation of the uncertainty surrounding that centre, providing a distinction between forecasting the centre vs forecasting the tails. The OU component determines

the expected mean-reverting path while the Student's t innovation determines how uncertainty is being distributed around that path. This makes modelling oil time spreads more realistic where periods of otherwise stable mean reversion could be interrupted by brief periods of large market movements.

2.3. Markov Regime Switching OU (with Student's t)

Oil markets exhibit vastly different behaviors under different market conditions; such as inventory shortages, inventory gluts or geopolitical shocks/high volatility crisis periods. Instead of relying on a single heavy-tailed error distribution ν with a single state to absorb sudden market shocks, a multi-state Gaussian HMM is adopted to model fundamentally different market conditions.

Building on the Student's t extension in the previous section, a further extension is to model the time spreads as a three-state Markov-switching OU process, where conditional on latent regime S_t , the spread is assumed to follow an OU diffusion with regime-dependent parameters $(\theta_i, \mu_i, \sigma_i)$. A single regime OU process forms a compromise trying to fit negative spreads, positive spreads and extreme spreads simultaneously, resulting in a poor average. By isolating them into distinct regimes, the HMM-OU is able to dynamically switch its conditional mean, capturing the non-linear behavior that a single-state model misses. When the market undergoes regime shifts (such as geopolitical events) that creates multimodal distributions, a single regime OU will need a long time to drag up its single mean value slowly, whereas a regime-switching model reacts faster by switching probabilities to a new state whose mean is anchored higher.

The model assumes that the spread probabilistically belongs to one of the three latent states:

1. *Contango*: This regime corresponds to periods of oversupply where spreads fall into negative territory (current month oil price is lower than one month ahead). This regime is bounded below 0 to capture surplus inventory regimes, and mean reversion is mainly driven by the Theory of Storage and cost of carry (Brennan (1958); Working (1949)). When future prices becomes relatively much higher than current prices, traders can arbitrage by buying and storing the oil. This puts upward price pressure on spot prices and downward pressure on futures, thus driving the spreads back to its fair value/carry relationship.

2. *Backwardation*: Mild backwardation represents the dominant regime in history and indicates a physically tight market. This regime is driven by convenience yield; the economic benefit of physically possessing the commodity now (Kaldor (1939); Schwartz (1997)). In this market state, the current month oil prices are slightly higher than one month ahead as supply is scarce today relative to expected future availability and hence refiners value immediate crude oil availability (high convenience yield). Mean reversion in this regime is driven by either an increased oil production by producers, or a reduction in demand.

3. *Crisis*: This regime is characterized by acute physical scarcity, regional supply shocks, geopolitical conflicts or bottlenecks such as oil pipeline constraints. Spreads spike up to extreme positive values and move far away from its fundamental value. This regime drives the long right tail in the distribution. As highlighted by Yuan and Kwon (2025), modelling this regime independently prevents structural tail events from contaminating baseline parameter estimates. The 3 regimes can be seen in section 4.1 figure 2.

HMM-OU setup

Conditional on being in regime $S_t = k$ at time t , the transition equation over time step h is given by:

$$x_{t+h} = \mu_k + (x_t - \mu_k)e^{-\theta_k h} + s_k \cdot T_\nu \quad (19)$$

Where:

x_t : The state variable (oil time spread) at time t .

$S_t \in \{1, 2, \dots, K\}$: A latent first-order Markov chain governing the active market regime ($K = 3$ for Contango, Normal, Crisis).

$\theta_k > 0$: The rate of mean reversion for regime k .

μ_k : The long-term mean equilibrium level for regime k .

$T_\nu \sim t(\nu)$: A standard Student's t - t random variable with ν degrees of freedom.

s_k : The regime-specific scale parameter, which links the continuous-time OU volatility σ_k and mean-reversion speed θ_k to the Student's t - t scale over time step h :

$$s_k = \sqrt{\left(\frac{\sigma_k^2}{2\theta_k} (1 - e^{-2\theta_k h})\right) \cdot \frac{\nu}{\nu - 2}} \quad (20)$$

The conditional expectation $m_{k,t}$ of the OU process in regime k over a time step h is given by:

$$m_{k,t} = \mu_k + (x_t - \mu_k)e^{-\theta_k h} \quad (21)$$

The emission density (observation likelihood given a regime) of observing x_{t+h} given x_t in regime k follows a Student's t distribution:

$$f_{\text{Student's } t}(x_{t+h} \mid x_t, \theta_k, \mu_k, \sigma_k) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right) s_k \sqrt{\pi\nu}} \left[1 + \frac{1}{\nu} \left(\frac{x_{t+h} - m_{k,t}}{s_k}\right)^2\right]^{-\frac{\nu+1}{2}} \quad (22)$$

Where:

$m_{k,t} = \mu_k + (x_t - \mu_k)e^{-\theta_k h}$ is the conditional mean location.

$\Gamma(z) = \int_0^\infty e^{-u} u^{z-1} du$ is the standard Gamma function, which normalizes the heavy tails for ν degrees of freedom

ν : The degrees of freedom parameter, controlling the kurtosis (fatness of the tails) of the Student's t - t distribution

s_k is defined above in the discrete transition equation

Regime transition dynamics

Regime dynamics are governed by a first-order Markov chain with transition matrix P where each element p_{ij} is the conditional probability of transitioning from regime i at time $t - 1$ to regime j and time t :

$p_{ij} = \mathbb{P}(S_t = j \mid S_{t-1} = i)$, with $i, j \in \{1, \dots, K\}$ and $\sum_{j=1}^K p_{ij} = 1 \quad \forall i$ because the market must be in some state tomorrow, the transition probabilities in each row must sum to 1.

As the regimes are latent, the model parameters are estimated using an iterative Expectation-Maximization algorithm, where the filtered regime probabilities calculated in the E-step below are subsequently used as weights in the MLE of the regime-specific parameters in the M-step.

Regime probabilities estimation - E-step

For a given time step t , $\alpha_t(k)$ represents the posterior probability of the market being in latent regime k at time t , given all historical observations available up to time t ($x_1, x_2, \dots, x_t$), defined as:

$$\alpha_t(k) \equiv P(S_t = k \mid x_{1:t})$$

Using Bayes' theorem, the regime probabilities are recursively updated based on the normalized product of the prior ($\alpha_{t|t-1} = \alpha_{t-1}P$) and the Student's t emission density given in equation (22) above to give the filtered posterior probability :

$$\alpha_t(k) = \frac{\alpha_{t|t-1}(k) \cdot f_{\text{Student's t}}(x_t \mid x_{t-h}, \theta_k, \mu_k, \sigma_k)}{\sum_{j=1}^K \alpha_{t|t-1}(j) \cdot f_{\text{Student's t}}(x_t \mid x_{t-h}, \theta_j, \mu_j, \sigma_j)} \quad (23)$$

Weighted MLE / MAP Parameter Optimization - M step part 1:

While [Cappé et al. \(2005\)](#) provides a solid foundational framework for Bayesian inference in hidden Markov models, [Green \(1990\)](#) extends the EM algorithm to integrate penalty functions into the M-step of the EM via penalized likelihood estimation, allowing structural economic information of regimes to be incorporated directly into optimization via informative priors and smooth boundary penalties.

During the M-step optimization, the parameters $\Theta_k = (\theta_k, \mu_k, \sigma_k)$ for each regime k are updated by minimizing the penalized negative log-posterior in the weighted Maximum A Posteriori (MAP) objective (explained more in section 3.2 point 3 below), using the posterior regime weights $w_t = \alpha_t(k)$ from the E-step above:

$$\mathcal{L}_{\text{MAP}}(\Theta_k) = - \sum_{t=0}^T \alpha_t(k) \ln f_{\text{Student's t}}(x_{t+h} \mid x_t, \Theta_k) - \ln p_k(\Theta_k) \quad (24)$$

where the first term is the base log-likelihood term from the emission density in equation (22) and the second term is a prior distribution $p_k(\Theta_k)$ where the MAP regularizes the objective function by introducing half-Normal priors which act as soft penalties on economically impossible parameter zones for each regime. $\alpha_t(k)$ are the regime weights from the E-step above.

For an upper boundary b , the half-normal prior density evaluated past the boundary is:

$$\phi(z; b, a^2) = \frac{1}{a\sqrt{2\pi}} \exp\left(-\frac{(z-b)^2}{2a^2}\right)$$

Taking the log of the density gives:

$$\ln \phi(z; b, a^2) = -\frac{1}{2} \ln(2\pi a^2) - \frac{(z-b)^2}{2a^2}$$

Upper boundary penalty:

$$p_{\text{upper}}(z; b, a) = \begin{cases} \frac{(z-b)^2}{2a^2}, & z > b \\ 0, & z \leq b \end{cases}$$

Lower boundary penalty:

$$p_{\text{lower}}(z; b, a) = \begin{cases} \frac{(b-z)^2}{2a^2}, & z < b \\ 0, & z \geq b \end{cases}$$

where z is the parameter value, b is the boundary threshold (upper or lower) and a is the penalty scale that governs the steepness of the penalty. When the estimated parameter z lies inside the allowed domain ($z \leq b$ for upper bounds or $z \geq b$ for lower bounds), the penalty function evaluates to 0. When the bounds are violated, the penalty term is activated as a squared distance penalty.

This is solved numerically via bounded quasi-Newton optimization using L-BFGS-B, established by [Zhu et al. \(1997\)](#) and [Byrd et al. \(1995\)](#), incorporating Half-Normal boundary priors ([Gelman \(2006\)](#)). The L-BFGS-B is a quasi-Newton optimization method which calculates the curvature (hessian matrix) of the log-likelihood function.

Transition matrix update - M step part 2:

The empirical transition probabilities $P(i, j)$ are updated using normalized transitions between states:

$$P(i, j) = \frac{\sum_{t=1}^{T-1} \alpha_t(i) P_{ij} f_{\text{Student's } t}(x_{t+h} | x_t, \Theta_j)}{\sum_{j=1}^K \sum_{t=1}^{T-1} \alpha_t(i) P_{ij} f_{\text{Student's } t}(x_{t+h} | x_t, \Theta_j)} \quad (25)$$

Forecasting:

To project h steps ahead (forecast horizon h) from the current time T , the forecast blends independent predictions from each regime into a mixture distribution.

Projected Mixture Weights (h -step transition):

$$w^{(h)} = \alpha_T P^h$$

Regime-Specific h -step Parameters: Substitute forecast horizon h into the OU conditional mean ($m_{k,h}$) and scale ($s_{k,h}$).

Mixture Expected Value:

$$\bar{m} \equiv \mathbb{E}[X_{t+h}] = \sum_{k=1}^K w_k^{(h)} m_{k,h}$$

Variance:

As the future latent state S_{T+h} is uncertain, the total variance of the mixture distribution is derived through the law of total variance and can be decomposed into within-regime uncertainty and between-regime uncertainty:

$$\text{Var}(X_{T+h}) = \mathbb{E}[\text{Var}(X_{T+h} \mid S_{T+h})] + \text{Var}(\mathbb{E}[X_{T+h} \mid S_{T+h}])$$

$$V_{\text{mixture}} = \underbrace{\sum_{k=0}^K w_k^{(h)} v_{k,h}}_{\text{Within-regime uncertainty}} + \underbrace{\sum_{k=0}^K w_k^{(h)} (m_{k,h} - \bar{m})^2}_{\text{Between-regime uncertainty}}$$

The first term captures the price volatility within each specific regime k , adjusted for Student's t heavy tails :

$$v_{k,h} = \left(\frac{\sigma_k^2}{2\theta_k} (1 - e^{-2\theta_k h}) \right) \cdot \frac{\nu}{\nu - 2}$$

The second term captures the dispersion of the expected regime means $m_{k,h}$.

2.4. Evaluation Metrics

Table 1.: Summary of Model Performance Metrics

Metric	Description	Formula
Root Mean Squared Error (RMSE)	RMSE measures the average magnitude of the forecasting error. It is compared against a naive random walk model (where the next period's forecast equals the current period's realization). This random walk assumption is a standard null hypothesis in financial forecasting, rooted in the Efficient Market Hypothesis (Fama (1970)). A lower OU RMSE implies that a mean-reverting process is able to systematically outperform the null hypothesis that future spread changes are inherently unpredictable.	$\sqrt{\frac{1}{N} \sum_{t=1}^N (X_t - \hat{X}_t)^2}$
Mean Absolute Error (MAE)	MAE measures the average absolute magnitude of the errors. It is much less sensitive to extreme outlier observations than the RMSE as it does not square the errors (linear penalty)	$\frac{1}{N} \sum_{t=1}^N X_t - \hat{X}_t $
Forecast Bias	Bias evaluates the average direction of forecast errors. A positive bias indicates systematic underprediction, whereas a negative bias indicates systematic overprediction.	$\frac{1}{N} \sum_{t=1}^N (X_t - \hat{X}_t)$
Directional Accuracy	Evaluates how well the model predicted the actual direction of future spread movements. Computes the percentage of times the model correctly predicted the actual direction of future spread movements using an indicator function 1 .	$\frac{1}{N} \sum_{t=1}^N \mathbf{1}_{\{\text{sgn}(\Delta X_t) \text{sgn}(\Delta \hat{X}_t)\}}$
Forecast Coverage	Coverage measures the percentage of realized observations that successfully fall within the OU model's 95% confidence interval bounds.	$\frac{1}{N} \sum_{t=1}^N \mathbf{1}_{\{L_t \leq X_t \leq U_t\}}$
Standardized Error Variance	Assesses the calibration of the model's volatility estimates. In a well-calibrated Gaussian framework, the variance of these standardized errors should ideally equal 1. A variance greater than 1 indicates that the actual forecast errors are consistently larger than the model's confidence intervals imply.	$z_t = \frac{X_t - \hat{X}_t}{\sigma_{\text{pred},t}}$ $\frac{1}{N-1} \sum_{t=1}^N (z_t - \bar{z})^2$
Out-of-Sample Density	Besides just evaluating point estimates, this metric assesses how accurately the predicted probability density f_t matched the actual realized market data X_t , evaluated using the Average Predictive Log-Score, where f_t is the forecasted probability density function evaluated at the actual realized observation X_t .	$\frac{1}{N} \sum_{t=1}^N \ln f_t(X_t)$

2.5. Programming the model

While there are many open-source Hidden Markov Model libraries available in computational packages such as [Schreiber \(2018\)](#), [Seabold and Perktold \(2010\)](#) and [hmmlearn developers \(2026\)](#), these ready-made solutions are more suited for generic classifications and standard time series tasks. The specific structural aspects of modelling Brent crude time spreads necessitate writing a full implementation from scratch. Open source HMM libraries default to Gaussian assumptions for emission probabilities, with no option to switch to Student's *t*. These packages also do not allow for custom constraints or penalties to be placed during the gradient descent, which would allow the statistical regimes to map to the physical economics of the oil market. This economic hierarchy prevents the 'label' switching problem inherent to EM, as the likelihood function is invariant to permutations of the regime labels. The likelihood surface of a multi-regime HMM-OU model is highly non-convex. General packages rely on static k-means clustering or uniform random initialization to seed the optimizers. These simplistic initialization methods result in gradient-based solvers to fail. The custom framework utilizes a probability-weighted initialization procedure within the EM loop. During each M-step iteration, the initial guess for the L-BFGS-B optimizer is analytically derived from the E step (forward probabilities). This ensures that the optimizer starts near the coordinates for that specific regime's conditional mean and volatility, avoiding flat gradients and bypasses local minima. With a custom-written package, it also facilitates the ability to potentially fit different models in different regimes, which would not be possible in a ready-made package.

3. Model design choices

3.1. Model architecture

1. **Number of regimes:** The number of regimes in this case is determined by domain knowledge of oil market structure as elaborated in section 2.3. In general, fitting more regimes typically yields higher in-sample log likelihood but is prone to overfitting as it captures market noise as 'mini regimes', while a single regime is inadequate to model real world scenarios, reflecting a bias-variance tradeoff.
2. **State transition matrix:** In the current implementation, the state transition matrix is re-estimated across each refit and adapts over time as the window slides. However, the transition probabilities are not explicitly modelled as a function of other real-time exogenous variables. An alternative is to use Time-Varying Transition Probability (TVTP) models pioneered in econometrics by [Diebold et al. \(1994\)](#); [Filardo \(1994\)](#), which allows transition parameters to respond dynamically to exogenous economic drivers. In the context of oil markets, exogenous variables could include inventory levels, refinery utilization rates that could improve responsiveness of regime switches. While the existing setup balances structural adaptability without needing complex modelling, it reduces the risk of overfitting, misspecification and much higher computational complexity. As the matrix is updated during each window to adapt to any major structural market shifts, it is computationally efficient and reduces overfitting noise, although it does not change as dynamically as in the alternative of varying the matrix at every single time step.

3.2. Parameter constraints and initialization

1. **Initialization of transition matrix:** Oil markets are anchored by physical fundamentals such as storage costs, inventory cycles and physical bottlenecks. As such, a random initialization would be inappropriate as it results in the EM getting stuck in poor local maxima. Hence, domain-guided initialization is used where the diagonals are sticky, reflecting how commodity regimes typically behave and to ensure that the optimizer does not converge on a model that rapidly oscillates between states frequently.
2. **Initialization of OU parameters for each regime:** The likelihood surface of a HMM is highly non convex and poor initialization (etc all regime same u), the optimizer will fail to separate them and lead to regime collapse. Instead, they are being initialized with data percentiles to force the optimizer to explore distinct states. Although in an expanding window set up, training dataset grows along the way and the effect of initialization lessens over iterations.
3. **Parameter bounds for each regime:** Instead of allowing the the model to optimize parameters unrestrictedly, which leads to economically uninterpretable regimes, the model imposes structural domain priors to reflect the physical realities of the commodity. [Frühwirth-Schnatter \(2006\)](#) established that unconstrained HMMs are fundamentally unidentifiable, with the optimizer randomly switching labels. The only way to perform cluster-specific inference is to apply informative priors or structural constraints based on expert domain knowledge. It details the necessity of using MAP estimation to prevent empty component collapse and explains using ordered or truncated priors to solve the label switching problem inherent to EM. [Hamilton and Susmel \(1994\)](#) justifies why economic thresholds are necessary to separate structural mean shifts from transient noise. [Engel and Hamilton \(1990\)](#) attempted to model exchange rates using unconstrained HMMs and the models frequently failed to align with known economic cycles. The authors structurally constrained the transition probabilities and means to fix this issue and respect the domain knowledge of ‘appreciation’ vs ‘depreciation’ regimes. [Rydén et al. \(1998\)](#) proved that standard, unconstrained HMMs fail to capture variance clustering of financial time series and the model could end up creating a state with infinite variance to absorb all shocks

Stability of regimes: Given that some regimes in some rolling windows could have very sparse or no data points, the optimizer could shrink the variance toward zero and resulting in the likelihood exploding to infinity (singularity). One solution to prevent regimes from such a collapse during unpopulated states is to use Maximum A Posteriori (MAP) estimation, where the conjugate prior for the variance is the Inverse-Gamma (IG) distribution, which ensures the volatility can never reach absolute zero or infinity. [Fraley and Raftery \(2007\)](#) established the standard framework for this technique, demonstrating that applying conjugate priors (such as IG for variance) regularizes the EM algorithm and prevents these singularities.

Regime label switching: Unconstrained HMM models inherently suffer from ‘label switching’ across optimization runs as the likelihood function is invariant to permutations of the regime labels, allowing the optimizer to arbitrarily exchange the identities of the latent regimes between iterations ([McLachlan and Peel \(2000\)](#)) (i.e. state 1 could be contango

regime in one run and crisis regime in another). This poses an issue as it complicates the interpretation of the estimated regimes due to inconsistent regime classification over time.

A half-Normal penalty function was constructed, ensuring a continuous and differentiable approximation of a Truncated Normal prior (as the L-BFGS-B optimizer requires continuous gradients to navigate constraints). This is used in place of another alternative considered, which is to impose strict ordering constraints via Truncated Normal Priors that enforce an economic hierarchy such as $\mu_0 < \mu_1 < \mu_2$. This option does not allow the solver to compute gradients smoothly up to the empirical regime boundaries.

Variance constraints for regime identification: Without having explicit variance constraints, regime-switching models frequently misclassify genuine structural mean shifts as transient heteroskedastic shocks (Hamilton (1989)). Hence, dynamic volatility ceilings are imposed based on the empirical standard deviation of the rolling window. By restricting infinite variance, the optimizer is able to model extreme price dislocations by shifting the long-run mean.

To address the above issues, a custom penalized log-likelihood objective function was implemented using a Maximum A Posteriori (MAP) framework. This helps to enforce economic reality of the regimes and to ensure solver stability.

3.3. Optimization mechanics

1. **Number of EM iterations:** The number of iterations determines how many cycles the EM loop runs and it affects the stability, accuracy and computational efficiency of the rolling backtest. On one hand, too few iterations leads to underfitting and parameters may fail to converge while too many iterations could lead to overfitting and increase computational time on very trivial likelihood improvements.
2. **Choice of optimizer:** The L-BFGS-B algorithm by Zhu et al. (1997) and Byrd et al. (1995) was used for optimization, where it approximates the inverse Hessian matrix using gradients. This ensures rapid convergence during refits without the computational drag of other derivative-free optimizers such as Nelder-Mead. The L-BFGS-B also allows for custom priors/bound constraints to be used without breaking convergence, such as the regime specific constraints described above.
3. **Inertia:** An inertia parameter was incorporated into the backtest loop to act as a temporal smoothing mechanism for the regime probability weights across time steps. This prevents the weights from jumping violently from week to week due to capturing short term market noise. The parameter determines how much of previous period's weights are carried forward to the next. A high inertia value makes the model much more stable and smooth but less responsive and creates lags and slower transitions, while a low inertia value reacts faster to market changes but is more prone to false positives and noisy regime classifications.

3.4. HMM EM psuedo code

Model Initialization: Before executing the EM loop, the system parameters and transition structures are initialized as follows:

The regime means are split by percentiles, where contango takes the 10th percentile, backwardation the median and crisis state taking the 90th percentile and scales initial volatilities σ_k relative to the overall sample difference variance (max standard deviation of the first-differenced data), representing the baseline day-to-day noise or volatility of the market and also bounded below to prevent division-by-zero.

The prior transition matrix is guided by an economic belief on how sticky each market regime is, with crisis being the least sticky state and the other two non-crisis regimes having higher probabilities of remaining in their states. An inertia parameter of 0.2 is also used to prevent erratic and unstable transition probabilities when sample sizes in tail regimes are small, by assigning 20% weight to the prior.

$$\begin{aligned}
 m_0 &\leftarrow \text{Percentile}(X, 10), & m_1 &\leftarrow \text{Median}(X), & m_2 &\leftarrow \text{Percentile}(X, 90) \\
 \sigma_{\text{overall}} &\leftarrow \max(\text{StdDev}(\Delta X), 10^{-4}) \\
 P_{\text{prior}} &\leftarrow \begin{bmatrix} 0.90 & 0.10 & 0.00 \\ 0.05 & 0.90 & 0.05 \\ 0.00 & 0.30 & 0.70 \end{bmatrix}, & \eta &\leftarrow 0.20
 \end{aligned}$$

Algorithm 1 HMM-OU Expectation-Maximization

Require: Time series $X = \{x_0, x_1, \dots, x_T\}$, time step h , regimes $K = 3$, iterations N_{iter}

Ensure: OU parameters $\{\theta_k\}$, transition probability matrix P , filtered state probabilities α

```

1: Initialize regime parameters  $\theta_k = (\kappa_k, \mu_k, \sigma_k)$  via data percentiles
2: Initialize transition matrix  $P \leftarrow P_{\text{prior}}$ , inertia  $\eta \leftarrow 0.20$ 
3: for  $iter = 1$  to  $N_{\text{iter}}$  do
    — E-Step: State Filtering —
4:    $\alpha, \mathcal{L} \leftarrow (X, \{\theta_k\}, P, h)$  ▷ Computes state weights  $\alpha_{t,k}$ 
    — M-Step Part 1: Regime Parameter Optimization (Weighted MAP) —
5:   for  $k = 1$  to  $K$  do
6:      $w \leftarrow \alpha[:, k]$  ▷ Extract posterior weight vector for regime  $k$ 
7:      $\theta_k \leftarrow (X, h, w, k)$  ▷ Parameter Updates  $\kappa_k, \mu_k, \sigma_k$ 
8:   end for
    — M-Step Part 2: Transition Matrix Re-estimation —
9:    $A_{\text{num}} \leftarrow \mathbf{0}_{K \times K}$ 
10:  for  $t = 0$  to  $T$  do
11:    Compute Student's  $t$ - $t$  emission probabilities  $b_i(X_{t+h} | X_t)$  using updated  $\{\theta_k\}$ 
12:    Update joint transition counts  $A_{\text{num}}$  using  $\alpha_t$ , emissions, and  $P$ 
13:  end for
14:  for  $i = 0$  to  $K$  do
15:     $P_{\text{raw}}[i] \leftarrow A_{\text{num}}[i] / \sum_j A_{\text{num}}[i, j]$ 
16:     $P[i] \leftarrow (1 - \eta)P_{\text{raw}}[i] + \eta P_{\text{prior}}[i]$  ▷ Inertia
17:  end for
18:  Normalize  $P$ 
19:  Sort  $\{\theta_k\}, P$ , and  $\alpha$  by long-term mean  $\mu_k$  ascending ▷ Prevent label indices switching
20: end for
21: return  $\{\theta_k\}, P, \alpha$ 

```

3.5. Backtesting/execution

A rolling/sliding window approach is adopted as the OU parameters do not remain static over the entire period of history over the past 16 years. Instead, a more valid and relevant assumption would be local stationarity, a concept formalized by [Dahlhaus \(1997\)](#). This allows the equilibrium price, mean-reversion speed and volatility to evolve over time, more accurately reflecting changing market dynamics. The windows are also overlapping as when the model steps forward to the next refit, it drops the single oldest week and adds the single newest week

Sequential out-of-sample backtesting procedure: At each time t , OU parameters are estimated using only a fixed specified historical lookback window available up to t to avoid look ahead bias. A $t+3$ forecast is generated and compared with the subsequently observed value. The procedure is repeated sequentially throughout the whole dataset and performance is evaluated.

The forecast represents the expected future spread level and can be interpreted as a weighted combination of the current spread level and the estimated long-run equilibrium mean level (in

equation 3). The model also generates a confidence interval around the expected future value, providing a measure of uncertainty around the forecast (equation 4).

1. **State inference:** The choice of using filtering (uses data up to time t) vs smoothing (uses full dataset up to time T). For backtesting, it uses filtering as smoothing uses the entire data to assign states and introduces look-ahead bias.
2. **Refit frequency:** Refitting every period ensures model operates on the latest parameters but increases computational time and risk of fitting to noise. A lower refit frequency speeds up backtesting and reduces overfitting to noise but leads to stale parameters. The model was refitted every week during the backtest and recalculates its parameters as soon as a new data point enters the window.
3. **Lookback period:** An important choice in modelling locally stationary time-series is the length of rolling window. The selection of an optimal window is governed by a bias-variance trade-off in the presence of structural breaks (Pesaran and Timmermann (2007)), where short windows exhibit high variance, reacting aggressively to localized noise and long windows suffers from structural bias and is slow to react to market changes as it contains more outdated information. Furthermore, Inoue et al. (2017) demonstrate that optimal window selection must account for time-varying parameters.

The lookback period determines the amount of historical window of data used to estimate the model parameters and the transition matrix before generating the out-of-sample forecasts (a lookback of 80 weeks implies 80 weeks of history is required before the first forecast can be made). A short lookback makes the model highly adaptive to regime switches but is highly unstable as parameter estimates become unstable and a single period of extreme volatility could distort the estimates. A long lookback is more robust and provides sufficient data points to reliably fit a 3-regime HMM without overfitting. However, a lookback that is too long leads to severe structural lag as the window spans multiple macro cycles and the model becomes too slow to adapt.

To determine the optimal lookback period for the model and to avoid in-sample overfitting associated with picking just one window to run the model, an out-of-sample walk-forward optimization loops was evaluated on several window lengths spanning 1 to 4 years to find a best fit, summarized in table 1 below.

The Walk-Forward Optimization revealed a clear seasonal dependency in the optimal estimation window, where the model performs worse when fractional-year lookback periods are used (1.5, 2.5 and 3.5 years). This happens as fractional windows represent seasonal imbalance and captures an unequal number of seasonal cycles (for instance, two winters but only one summer), structurally distorting the empirical quantiles used to anchor the regimes. If a 1.5-year rolling window contained two high-demand winters and one low-demand summer, the median is artificially dragged upwards. Full-year windows perform better as it accounts for the seasonality prevalent in commodity markets. The 104 week window was adopted as the optimal rolling window for the final model.

Table 2.: Out-of-sample walk-forward optimization results across different lookbacks

Lookback	Years	RMSE	Hit Rate (%)	Bias
52 weeks	1.0	0.4585	58.38	0.0052
78 weeks	1.5	0.4685	47.25	0.0013
104 weeks	2.0	0.4597	60.12	0.0052
130 weeks	2.5	0.4820	47.93	0.0117
156 weeks	3.0	0.4686	59.38	-0.0013
182 weeks	3.5	0.4686	59.38	-0.0013
208 weeks	4.0	0.4813	58.11	-0.0082

Note: RMSE is evaluated in physical dollar spread terms. Hit Rate measures the directional accuracy of the expected forecast.

4. Results

The dataset contains daily crude brent oil time spreads, from January 2010 to August 2026, resampled to weekly every Friday. A time series plot of the data can be found in the top panel of figure 8. The spread value (M1-M2) represents the difference in current month oil price and oil price one month ahead. A negative spread ($M1-M2 < 0$) indicates that current month price (M1) is lower than future month (M2), representing contango state. A positive spread ($M1-M2 > 0$) is the opposite, representing backwardated markets. A histogram of the empirical data can be found in figure 2.

4.1. In-sample diagnostics

Establishing local stationarity across windows

For each rolling estimation window of length W (104 weeks in this case), the three-period persistence is estimated using a first-order autoregressive (AR(1)) specification fitted via standard OLS regression:

$$X_t = c_3 + \phi_3 X_{t-3} + \epsilon_t \quad (26)$$

where c_3 is the intercept term, ϕ_3 is the three-period persistence coefficient and ϵ_t is the residual error term.

Under the exact discrete-time representation of an OU process,

$$X_t = \mu + (X_{t-3} - \mu)e^{-3\theta} + e_t \quad (27)$$

and the three-period persistent coefficient is $\phi_3 = e^{-3\theta}$ when mapped to the exact analytical discrete OU solution,

Within each rolling window, the OLS estimate of the three-period persistent coefficient is obtained from the closed-form solution:

$$\hat{\phi}_3 = \frac{\sum_{t=4}^W (S_{t-3} - \bar{S}_{-3})(S_t - \bar{S})}{\sum_{t=4}^W (S_{t-3} - \bar{S}_{-3})^2}$$

where $\bar{S}_{-3}$ and $\bar{S}$ denotes the sample means of the lagged series s_{t-3} and the series s_t

The evolution of $\hat{\phi}_3$ across windows are plotted in the top panel of figure 1 below. The persistent parameter ϕ indicates how much of three periods ago deviation from equilibrium survives the next period, and $0 < \hat{\phi}_{3,t} < 1$ in most windows supports local stationarity assumption, implying that the dynamics of oil spreads follow a bounded, mean-reverting process.

Since $\hat{\phi}_3 = e^{-3\hat{\theta}}$, the implied local mean reversion speed is

$$\hat{\theta} = -\frac{\ln(\hat{\phi}_3)}{3} \quad (28)$$

The corresponding OU half-life is thus

$$HL_t = -\frac{3 \ln(2)}{\ln(\hat{\phi}_3)}$$

This translates the multi-period persistence parameter $\hat{\phi}_3$ into an economically intuitive temporal metric - the local OU mean reversion speed θ , and half-life (HL_t), which measures the expected duration (in weeks) required for a shock to revert midway back to its equilibrium.

The mean reversion speed and half-life across the rolling windows are plotted in the second and third panel of figure 1 below. The relationship between the persistent coefficient and mean reversion speed is given by equation 28, where high persistent indicates slower mean reversion and lower persistence implies faster mean reversion and shorter half-life. In the limiting case where $\hat{\phi}_t \rightarrow 1.0$, it approaches a random walk and $\hat{\theta} = 0$ and the half-life is undefined as $HL_t \rightarrow \infty$ (as observed in the latest months of the series in figure 1 below), indicating non-stationary or structural market breakdowns.

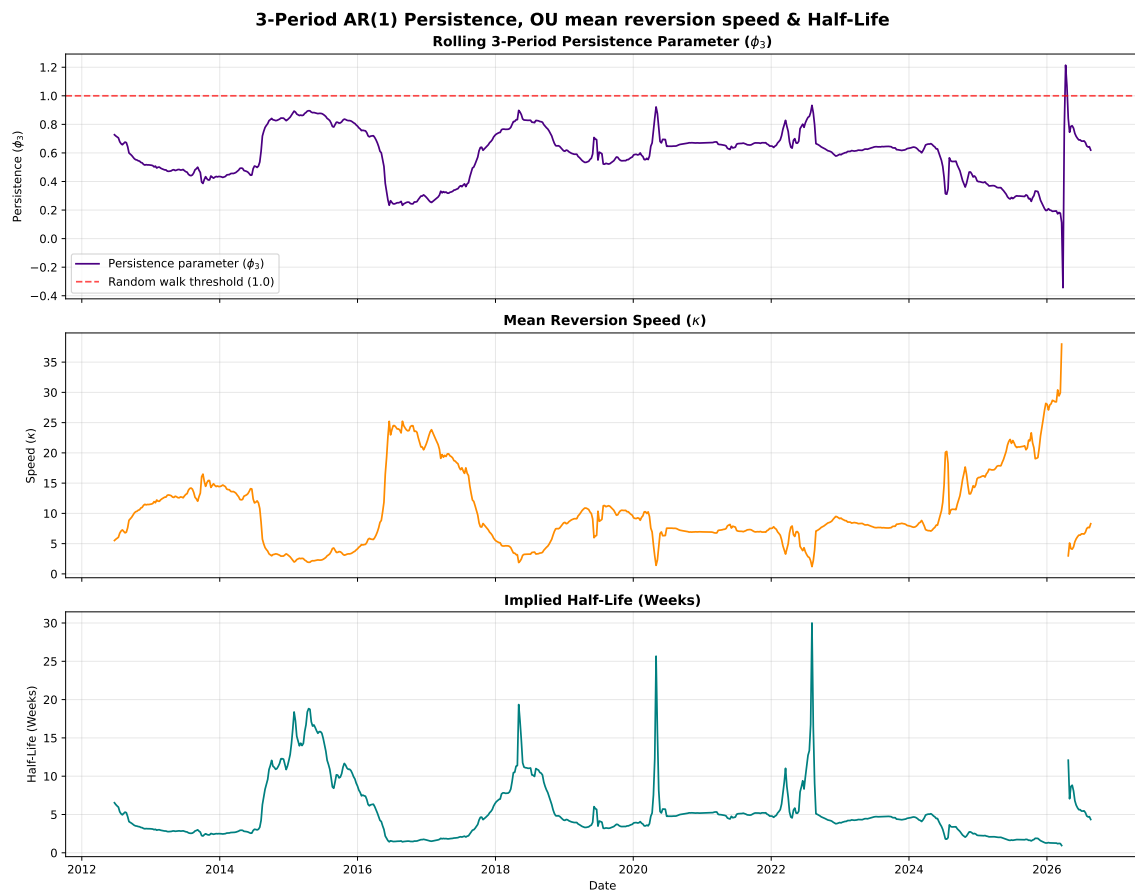


Figure 1.

The single-state OU density is plotted against the HMM regime densities in figure 2 below, overlaid on the full empirical histogram. It is evident that the single OU baseline struggles to find the true distributional shape of oil spreads as it forces a single global mean and variance across fundamentally distinct market states.

The backwardation regime (orange curve) has the highest peak and lowest variance, while the contango regime (green) reflects well-supplied inventory structures. The crisis regime (red) is characterized by an elevated long-run mean and a huge variance, representing unanchored market breakdowns caused by supply shocks or geopolitical events where standard physical arbitrage which ensures spreads stay close around 0 fails, generating the long right tail. The single OU (blue curve) regime has a peak that fails to capture the two peaks (backwardation and contango) and the whole curve is shifted right to compensate for the heavy right tails during crisis.

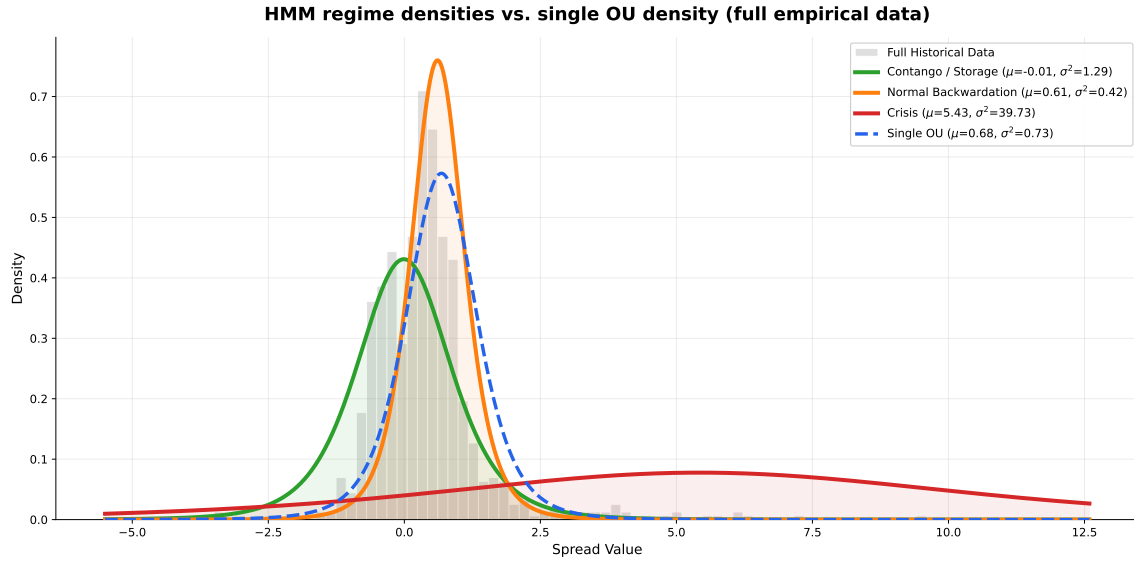


Figure 2.: For each individual regime density, the parameters are obtained from running the EM across the entire historical timeline and represents the steady-state of each regime vs a single-regime OU

4.2. Out-of-sample results

4.2.1. Student's t vs Gaussian OU model results

The Student's t OU model achieves a lower RMSE and MAE than the Gaussian OU. Both models maintain negligible forecast bias, with neither specification introducing systematic over or under forecasting tendencies over the backtest period. The mean standardized error for the Student's t (0.0878) is closer to the 0 than the Gaussian (0.17). The forecast bias is measured in the original units of data (\$/bbl) and checks if the model has a raw, unadjusted directional tendency to systematically over or under forecast, while the mean standardized error is normalized and evaluates if the error distribution is centered relative to the model's predicted uncertainty bounds, as each forecast error is divided by the period's predicted forecast standard deviation.

The Gaussian OU specification's standardized errors exhibit excessive dispersion ($\text{Std} = 1.559$), indicating severe miscalibration and variance is systematically underestimated, while the Student's t specification normalizes errors better ($\text{Std} = 1.2$).

Under the Gaussian assumption, extreme forecast errors (exceeding 2σ and 3σ) are elevated at 9.23% and 2.99% respectively. In contrast, the Student's t assumption suppresses these events down to 2.85% and 0.95%, as it has a wider interval to account for the probability of tail risks.

The empirical coverage with the Student's t addition (92.7%) is higher than the baseline Gaussian OU (90.5%) as the risk bounds are better quantified with the former's dynamic volatility cones.

The estimated degrees of freedom ($\bar{\nu} \approx 3.24$) is well below the Gaussian threshold ($\nu > 30$), validating that oil time spreads exhibit heavy tail behavior and invalidates standard Gaussian assumptions.

The data is also strongly right-skewed as supply shocks in an undersupplied, backwardated market are more pronounced than an oversupplied (contango) market. Furthermore, the downside of brent spreads are more constrained and storage costs provide a natural floor for prices while the upside risk during a crisis is unconstrained. The contango state is strictly bounded by the physical cost of storing oil (tank leasing, insurance, financing etc) and once the spread widens enough to cover these costs, arbitrageurs would buy physical oil, store it and sell the forward contract for a risk-free profit. This is also evident from the empirical data in figure 2.

Table 3.: Out-of-Sample Performance and Error Diagnostics: Gaussian OU vs. Student's t - t OU

Evaluation Metric	Gaussian OU	Student's t - t OU
Root Mean Squared Error (RMSE)	0.8512	0.8083
Mean Absolute Error (MAE)	0.4680	0.4353
Forecast Error Bias	0.0594	0.0689
Empirical Coverage (%)	90.50	92.67
Hit Rate (%)	59.84	59.43
Mean Standardized Error	0.1666	0.0878
Std. Dev. of Standardized Error	1.5591	1.1999
$ e > 2\sigma$ (%)	9.23	2.85
$ e > 3\sigma$ (%)	2.99	0.95
Forecast Error Skewness	2.1524	2.3983
Forecast Error Kurtosis	27.2932	26.2551
Avg. Degrees of Freedom ($\bar{\nu}$)	-	3.24

The figure 3 below compares the confidence intervals implied by the Gaussian and Student's t OU models. By construction, the Student's t intervals are wider than a standard Gaussian interval for finite ν . Consequently, the Student's t version anticipates extreme forecast errors more frequently and produces wider prediction intervals. The yellow dots below shows observations where it would fall outside a standard Gaussian CI but within Student's t CI due to heavier tails.

The Student's t model adapts its uncertainty according to the estimated tail behavior. During calm periods, ν is higher and the Student's t interval approaches the Gaussian interval. During crisis periods, ν falls and the Student's t CI widens more vs Gaussian, thus reducing the number of points falling outside the interval.

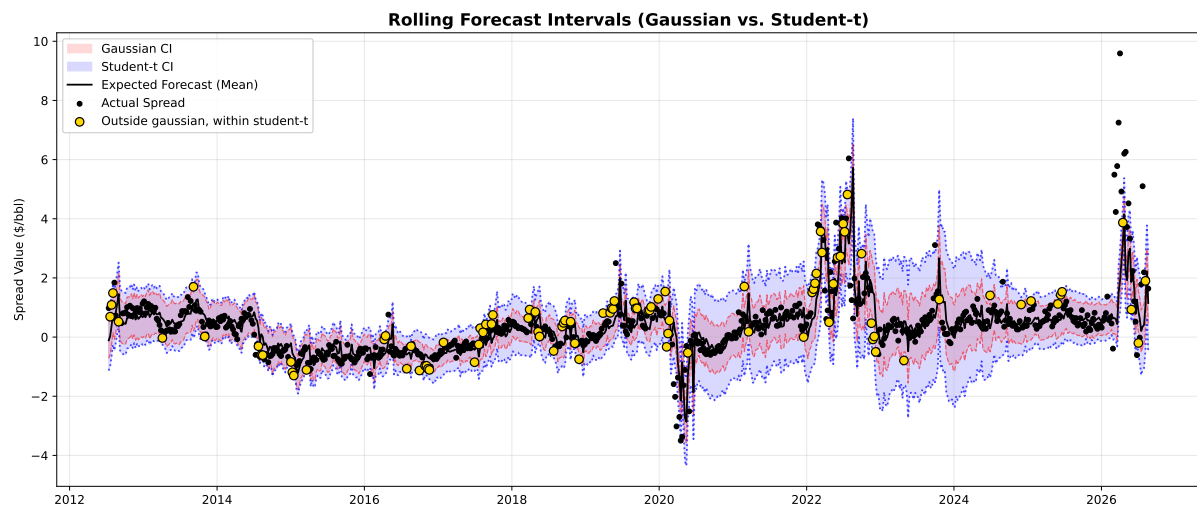


Figure 3.

Figures 4 below compares the rolling parameter estimates for the Gaussian vs Student's t OU model over the backtest period. In the second panel on the right, the Gaussian model's estimated mean saw a huge spike around 2022, as crisis shocks distort the optimizer in the Gaussian model to chase temporary market noise, while the Student's t estimated mean in the same crisis period remained anchored by downweighting extreme shocks.

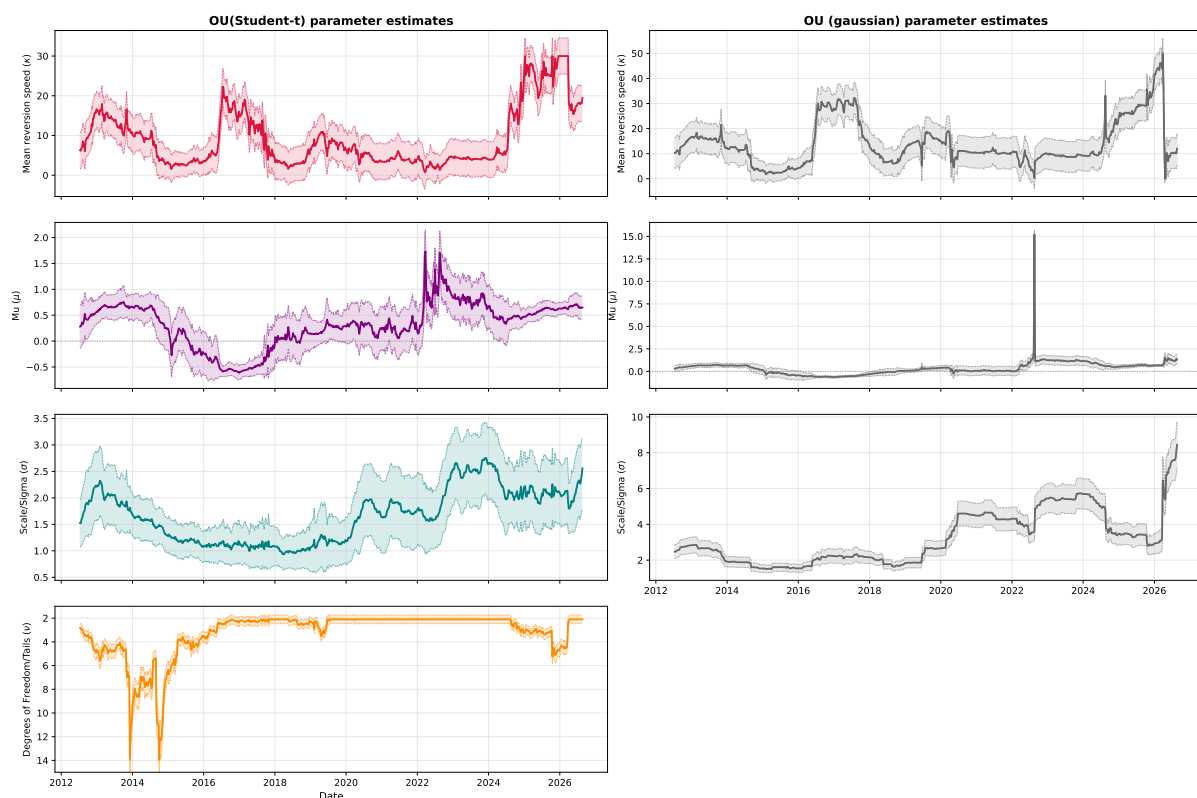


Figure 4.: Evolution of parameters across rolling windows

The two Q-Q plots in figure 5 plots the empirical standardized errors against theoretical quantiles under the Gaussian distribution and the Student's t distribution (parameterized by the

average estimated $\bar{\nu} \approx 3.24$. The left Gaussian plot flares out at both edges, showing that standardized errors contain far more extreme values than what a standard Gaussian allows for. In comparison, the scatter points on the right chart tracks the reference line more closely as it accounts for the heavier tails.

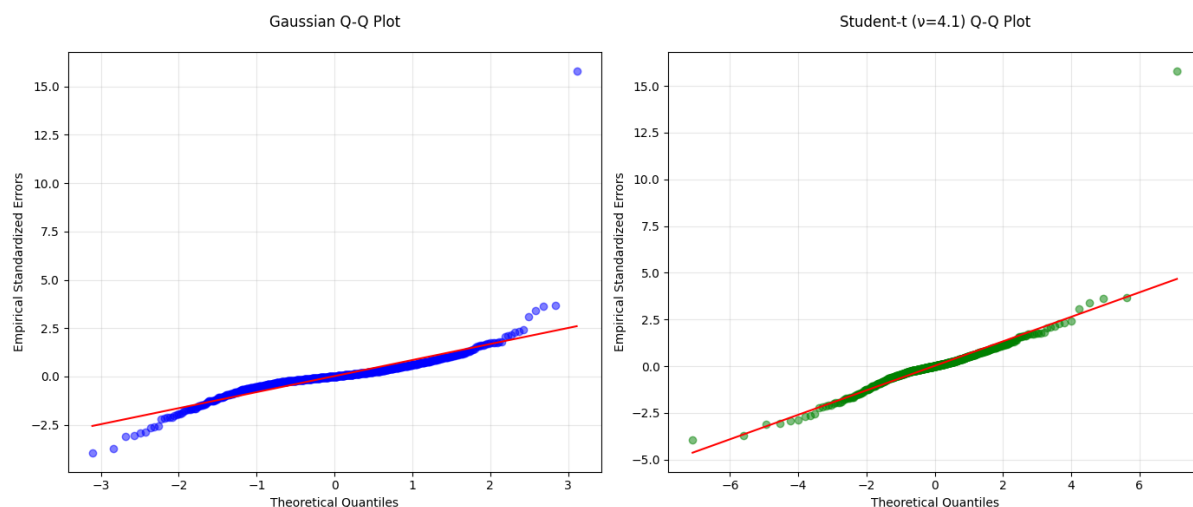


Figure 5.

4.2.2. Baseline Student's t OU

The baseline OU with Student's t model performance is split by Z-score (a measure of the number of standard deviations the current time spread is away from its mean), which allows for investigating how forecast accuracy depends on the magnitude of deviations from equilibrium. Table 4 shows that the OU model increasingly performs better (both RMSE and MAE) than the naive random walk forecast when Z-score is higher. This is due to naive forecast assuming next period's spread equals current period spread, while OU forecast assumes next period's spread equals current period spread plus some pullback factor towards equilibrium. Hence when the spread is only slightly displaced, the two sets of forecasts are similar while if the spread is highly displaced, the forecasts diverge. When the observations are close to the equilibrium, there is also less room for mean-reverting movement. Conversely, larger absolute Z-scores represent larger deviations from equilibrium and provides opportunities for OU process to improve on a random walk forecast.

The difference in MAE and RMSE is due to linear vs quadratic penalties. Using the MAE, an error of \$4 is penalized exactly twice as much as an error of \$2 but with the RMSE, an error of \$4 is penalized four times as much as an error of \$2. As such, for forecasts within the $|Z| \geq 3$ category, a handful of extremely large shocks are pulling the overall figures down. This emphasizes on the need to account for tail risks.

Table 4.: Out-of-Sample performance by Z-Score

Z-score	Periods	RMSE (\$/bbl)			MAE (\$/bbl)		
		OU	RW	Edge	OU	RW	Edge
$0 \leq Z < 1$	418	0.5483	0.5508	+0.44%	0.3403	0.3432	+0.84%
$1 \leq Z < 2$	216	0.5301	0.5442	+2.59%	0.3323	0.3532	+5.91%
$2 \leq Z < 3$	59	0.9228	1.0063	+8.30%	0.5621	0.6227	+9.73%
$ Z \geq 3$	44	2.3593	2.3684	+0.38%	1.6730	1.8484	+9.49%

4.2.3. Regime Switching HMM-OU results

Although for any given time period the HMM-OU model does not make a hard, deterministic classification of which regime the data point belongs to (instead it outputs a vector of continuous belief probabilities), for the purpose of evaluating model performance across different regimes, an argmax selection is applied on the continuous probability vector generated by the filter for that week and outputs the state with the highest numerical probability and assigns that week to that specific regime's basket. For instance, if for a given period, the filter outputs are $\alpha = [0.82, 0.15, 0.03]$, the argmax is index 0, which is regime 0 (contango regime). After every historical week in the backtest has had a regime label attached to it, regime-conditioned performance metrics are computed (tables 6-9).

Backwardation represents the dominant equilibrium state in oil market (with 73.1% occupancy), crisis states (8% occupancy) are rare and contango states (18.9%) happens occasionally when structural oversupply forces oil into storage (recorded in periods column in the tables below).

Across all periods, the regime switching HMM-OU model achieves the lowest RMSE and MAE, outperforming the single-regime OU by 4%. The largest gain of the HMM-OU model over the other models are concentrated in the crisis regime, with a +10.6% improvement over the single-regime OU. The improvement gains in the other two non-crisis regimes are minor. Although the crisis regime represents only 8% of the total time, the HMM-OU model is better able to forecast mean reversion over the single regime OU during times of crisis and reduces the chances of catastrophic losses.

Apart from having better mean forecast accuracy, the HMM-OU model also achieves the highest 95% coverage. In the single-regime OU, the coverage in the crisis regime is only 76%, indicating a severe underestimation of tail risks, whereas the HMM-OU maintains a more robust risk bound with 93.2% coverage, matching the 95% target more closely.

In the tables below, total number of periods in backtest = Total number of periods in dataset - lookback period (as the initial lookback window cannot be used as it is used to calibrate the first batch of OU parameters) - number of forecast steps ahead

Table 5.: Out-of-Sample Forecast Accuracy (HMM-OU vs. Single OU)

Market Regime	Periods	RMSE (\$/bbl)			MAE (\$/bbl)			Coverage(%)	
		HMM-OU	Single OU	Edge (%)	HMM-OU	Single OU	Edge (%)	HMM-OU	Single OU
Contango / Storage	139	0.4142	0.4161	+0.45%	0.2751	0.2720	-1.14%	98.5	98.5
Normal Backwardation	539	0.7822	0.7946	+1.57%	0.4255	0.4374	+2.72%	96.7	94.6
Crisis	59	1.4025	1.5683	+10.57%	0.8791	0.9421	+6.69%	93.2	76.3
All Periods	737	0.7983	0.8315	+3.99%	0.4335	0.4466	+2.93%	96.7	93.9

Table 6.: Out-of-Sample Forecast Accuracy(HMM-OU vs. Random walk)

Market Regime	Periods	Hit rate	RMSE (\$/bbl)			MAE (\$/bbl)			Coverage(%)	
			HMM-OU	RW	Edge (%)	HMM-OU	RW	Edge (%)	HMM-OU	RW
Contango / Storage	139	66%	0.4142	0.4160	+0.41%	0.2751	0.2836	+3.01%	98.5	92.1
Normal Backwardation	539	57%	0.7822	0.7909	+1.11%	0.4255	0.4382	+2.89%	96.7	92.0
Crisis	59	72%	1.4025	1.5188	+7.65%	0.8791	1.0541	+16.60%	93.2	89.8
All Periods	737	61%	0.7983	0.8215	+2.82%	0.4335	0.4584	+5.43%	96.7	91.9

As an additional benchmark, the HMM-OU model is compared against a 4-week Simple Moving Average (SMA),

$$SMA_t = \frac{1}{n} \sum_{i=0}^{n-1} P_{t-i}$$

The SMA is a momentum indicator which captures local drift and adapts to the recent local level of the market.

Table 7.: Out-of-Sample Forecast Accuracy (HMM-OU vs. 4-Week SMA)

Market Regime	Periods	RMSE (\$/bbl)			MAE (\$/bbl)			Coverage (%)	
		HMM	SMA	Edge	HMM	SMA	Edge	HMM	SMA
Contango / Storage	139	0.4142	0.4728	+12.38%	0.2751	0.3176	+13.38%	98.6	92.1
Normal Backwardation	539	0.7822	0.8755	+10.66%	0.4255	0.5115	+16.81%	96.7	91.8
Crisis	59	1.4025	1.9100	+26.57%	0.8791	1.3199	+33.40%	93.2	89.8
All Periods	737	0.7983	0.9459	+15.60%	0.4335	0.5397	+19.68%	96.7	91.7

Although the SMA is a standard baseline for local drift, it is slow to react as it applies an equal weightage to the previous prices in the window whereas an Exponential Moving Average (EMA) applies an exponentially decaying weight to historical data. The EMA at time t is:

$$EMA_t = \alpha P_t + (1 - \alpha) EMA_{t-1}$$

Where P_t is the current price and the smoothing factor α is defined by the chosen lookback window N :

$$\alpha = \frac{2}{N + 1}$$

For a 4-week EMA benchmark ($N = 4$), the smoothing factor is $\alpha = 0.40$, which means that 40% of the EMA's forecast is driven by the most recent observation, 24% by the period preceding that and so on.

The results in table 10 below shows that a mean reversion model is more suited for modelling crude oil time spreads than momentum indicators.

Table 8.: Out-of-Sample Forecast Accuracy (HMM-OU vs. 4-Week EMA)

Market Regime	Periods	RMSE (\$/bbl)			MAE (\$/bbl)			Coverage (\$/bbl)	
		HMM	EMA	Edge (%)	HMM	EMA	Edge (%)	HMM	EMA
Contango / Storage	139	0.4142	0.4335	+4.43%	0.2751	0.2890	+4.83%	98.6	88.5
Normal Backwardation	539	0.7822	0.8004	+2.27%	0.4255	0.4442	+4.21%	96.7	90.5
Crisis	59	1.4025	1.4602	+3.95%	0.8791	1.0312	+14.75%	93.2	93.2
All Periods	737	0.7983	0.8214	+2.81%	0.4335	0.4620	+6.17%	96.7	90.3

Besides relying only on point-forecast accuracies such as RMSE and MAE to evaluate the models, the out-of-sample performance is also evaluated using Log-Predictive Scores (LPS) aggregated cumulatively over time. The score is used to evaluate how well a probabilistic model predicts unseen data by measuring the log of the predicted probability density assigned to the actual observed outcome. [Gneiting and Raftery \(2007\)](#) provides the statistical justification for using proper scoring rules such as the cumulative log-predictive scores). [Amisano and Giacomini \(2007\)](#) establishes the framework for comparing density forecasts using relative log-score paths. [Geweke and Amisano \(2010\)](#) applies cumulative predictive likelihood scores directly to financial asset return models and demonstrated how tracking log-scores over time exposes structural model out-performances during market regimes.

As an illustrative example, if the actual spread for a period is 5, and the single-regime OU has its single mean far away from 5 and the model assigns near-zero density to this point, whereas the HMM-OU model assigns a higher density to this same point and as a result, gets a higher (less negative) log density score. These scores are then accumulated over time to determine if one model has a consistent structural edge over another (by having consistently assigning higher probabilities to realized market outcomes), over all rolling windows.

At each out-of-sample evaluation step, each model's predicted probability density function is evaluated at the value of the actual realized spread and the log of this value is summed up across all windows to obtain the cumulative log predictive score (top panel in figure 6). The HMM's cumulative forecasting edge, relative to the single regime OU is the difference in the two model's LPS (bottom panel in figure 6).

In the top panel of figure 5, the green trajectory belonging to the HMM-OU model consistently remains above or equal to the single OU model (red trajectory), establishing better probabilistic density coverage. The bottom panel of the same figure shows the difference (relative gap) in the accumulated log-scores over time. During times of major oil market shock (2020, 2022 and 2026), the edge of the HMM-OU versus single OU jumps as the latter suffers from high log-score penalties as its single regime parameters are unable to handle sudden regime shifts. In contrast, the HMM-OU model is able to reweight its latent state probabilities α_t towards the crisis regime and assigns higher probability density to the actual extreme outcome materialized. During calm

periods in history, the scores are similar and the gap between the two models is small as the need for regime separation is less pronounced as standard pricing dynamics dominate.

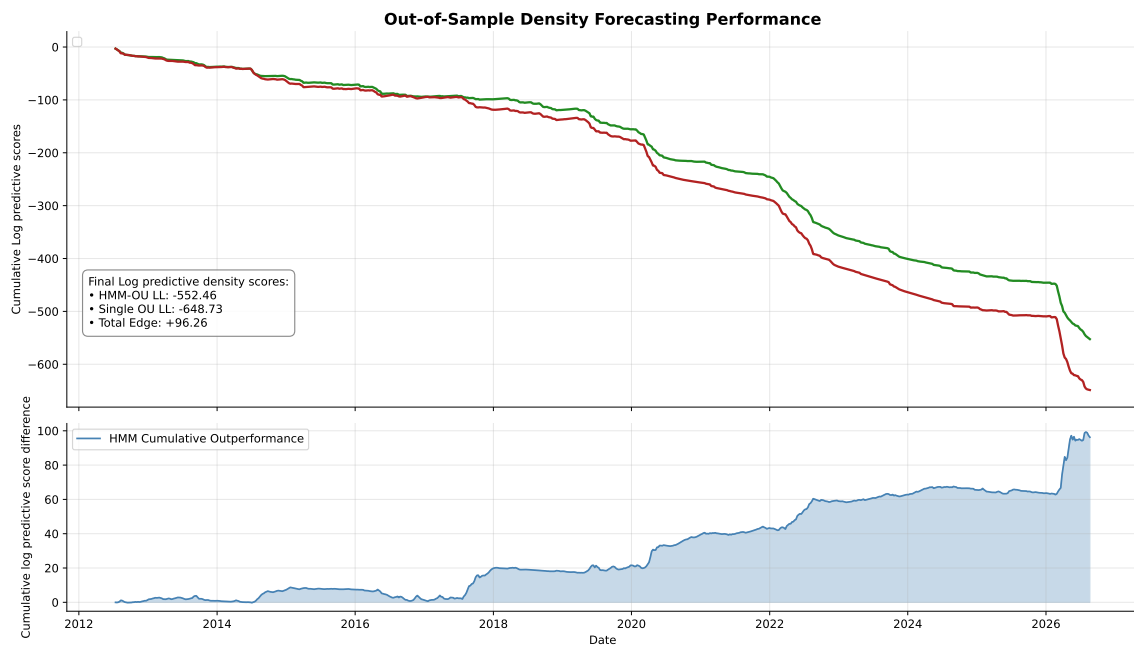


Figure 6.: The LL is negative as the value of the likelihood function is a decimal less than 1 and the log of it yields negative. The downward trend in the top panel is the result of summing over negative numbers across windows. Bottom panel shows the cumulative predictive info HMM gained over the single regime OU.

To assess if the estimated mean-reverting dynamics of the regime-switch HMM-OU remains stable over time, the OU parameters are tracked across the rolling windows for each regime. In the 3D cluster chart on the left panel of figure 7 below, each dot represents the fitted parameters of each regime across all historical rolling windows. It shows that the 3 regimes are spatially separated in the 3D parameter space. The right panel of figure 7 shows how each of the 3 OU parameters in each regime track over time, showing how the parameters drift across market cycles. Both charts taken together show that a single-regime OU is insufficient to capture oil market dynamics as mean reversion speed, long-run equilibrium mean and volatility are fundamentally regime-conditional as a single set of OU parameters would be inadequate to describe different market conditions.

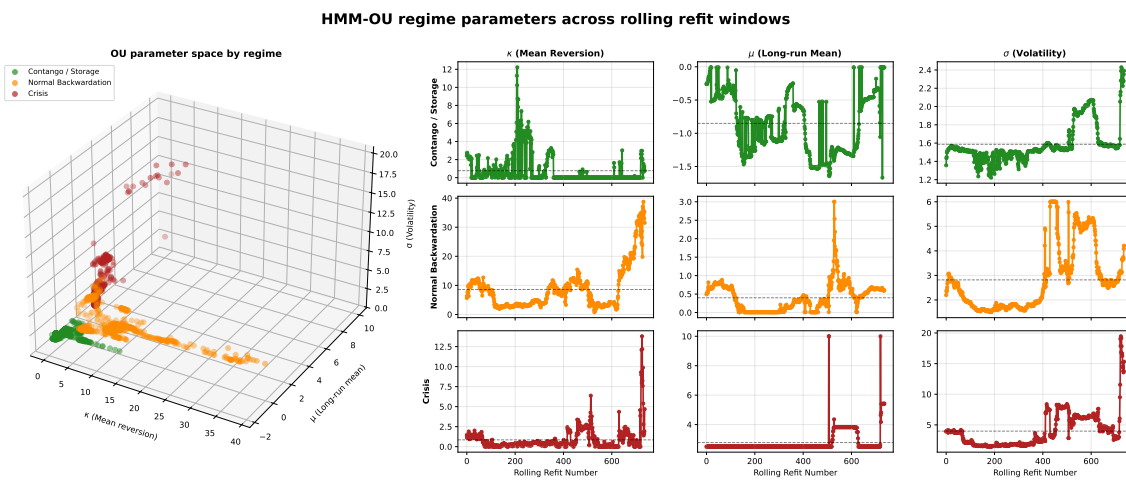


Figure 7.

The top panel of figure 8 shows the actual brent crude oil time spreads, the 3-step HMM-OU mixture forecast and its confidence interval. The colored bands in the bottom panel shows the stacked regime probabilities (α_t) for each time step, showing how the market shifts between structural states, with contango (green) and backwardation (green) dominating most of the time as these two regimes signify normal market conditions, while crisis (red) regime dominates only during extreme price moves.

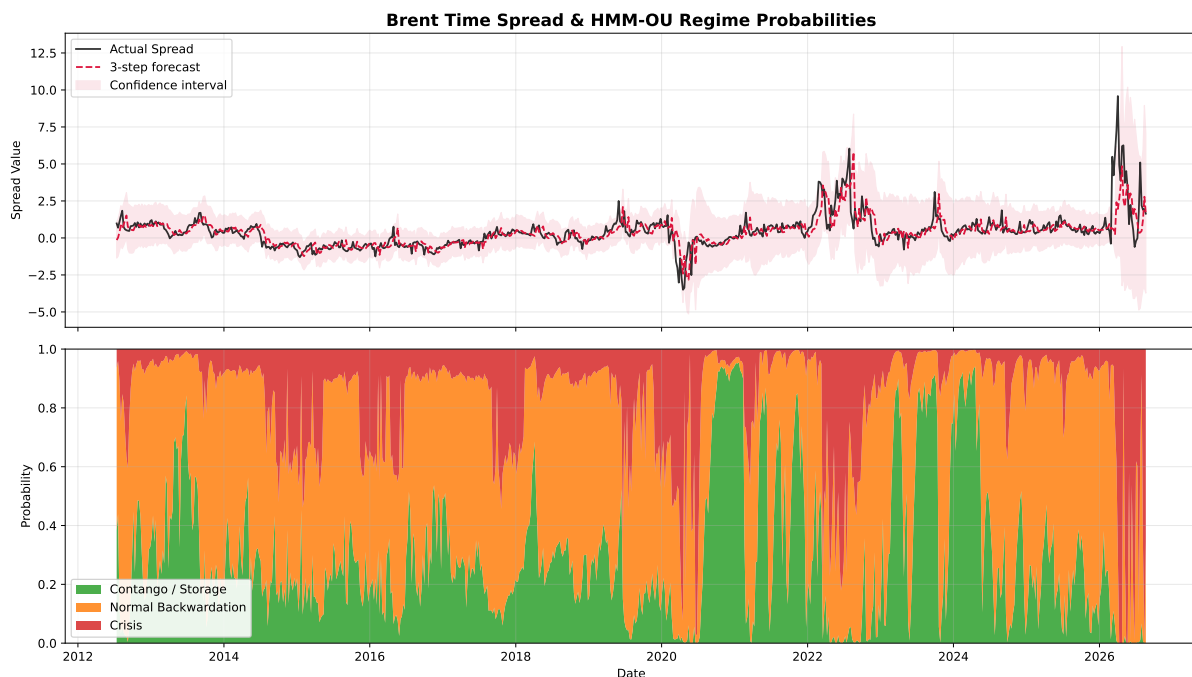


Figure 8.: Raw time series, forecasts and regime probabilities stack plot

Forecasts errors (figure 9) appear to be centered around zero most of the time, with no persistent systemic bias in either directions. However, occasional large errors are observed which coincides with abrupt market transitions where regime identification requires several periods to adapt.

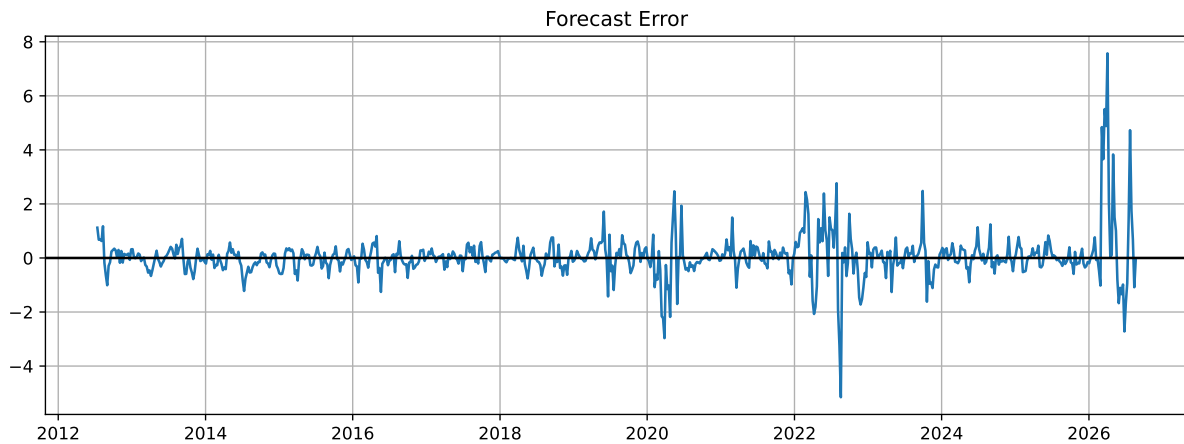


Figure 9.

The estimated transition matrix indicates that market regimes are persistent rather than transient (high value diagonals), especially during non-extreme market conditions (contango and backwardation regimes), whereas the crisis regime is relatively less persistent. In the final transition matrix, the crisis self-transition probability of 0.79 is higher than the historical average of 0.73 as the final estimation window contains much higher spikes relative to average conditions across the entire backtest.

The average expected duration (persistence) of regime i over the entire backtest is computed from the average transition matrix and plotted in figure 11 below:

$$\bar{D}_i = \frac{1}{1 - \bar{P}_{ii}}, \quad (29)$$

where $\bar{P}_{ii}$ is the i th diagonal element of the averaged transition matrix.

Figure 11 below shows the expected duration for each regime, derived from the estimated transition matrix from the backtest. It shows that the crisis regimes exhibit much lower persistence, relative to the other two stable market regimes. This suggests that shocks on oil time spreads do not last for long before it corrects back to equilibrium.

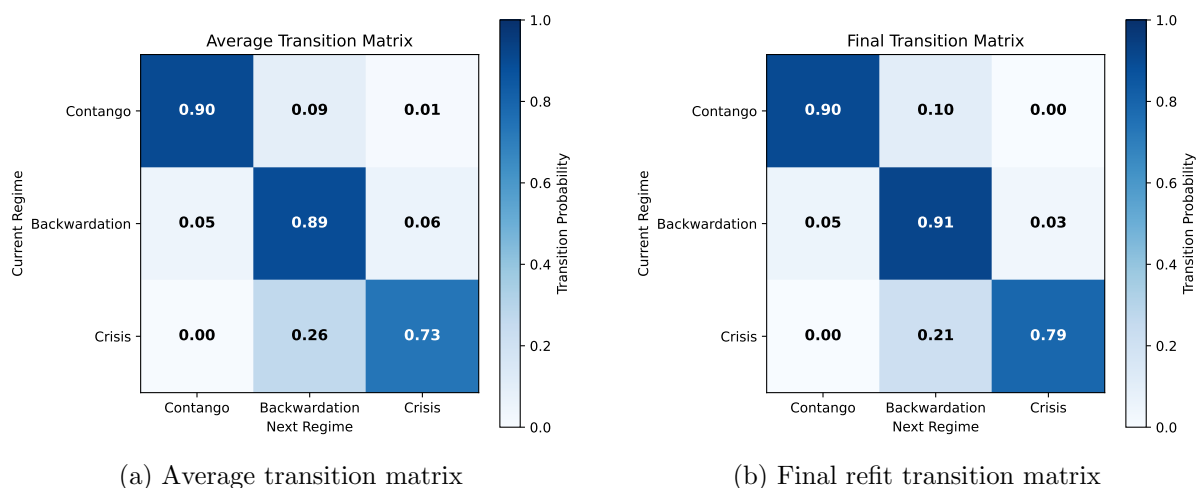


Figure 10.

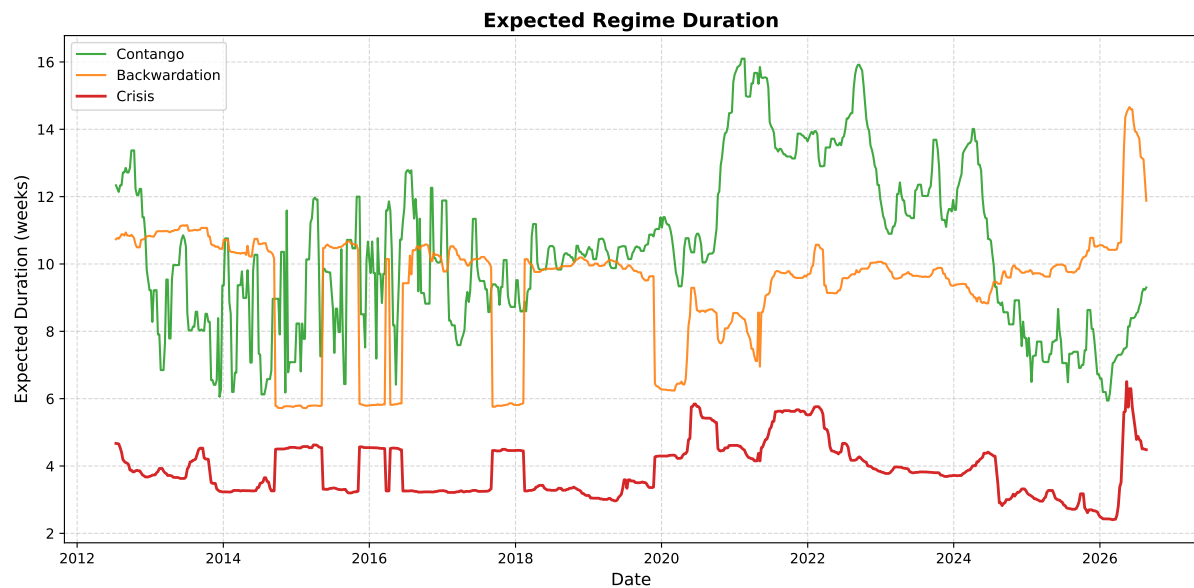


Figure 11.

5. Discussion

The out-of-sample results provide some evidence that using a three-regime HMM-OU framework improves both forecast performance and uncertainty quantification, relative to the single-regime OU benchmark and other naive benchmarks, although the magnitude of improvement varies across different regimes. The incremental gain in performance is mostly concentrated in the crisis regime as these periods are precisely why single-regime OU is likely to be misspecified, as a single OU process uses only one set of global parameters to accommodate both ordinary and extreme market conditions. The improvement of the HMM-OU is much more modest during normal market conditions.

Future work includes using time varying transition dynamics in the HMM model, where the transition probabilities do not remain static over time but instead, driven by exogenous market variables such as physical oil inventory levels, curve structure, etc which could sharpen the transition probabilities and improve forecast quality across all regimes. Although this has the potential for huge gains, it comes at the cost of much higher model complexity and possibly overfitting concerns. Another area of development would be to extend the M1-M2 spreads analysis by fitting a multi-factor OU model on the entire oil futures curve.

6. Endmatter

The crude oil time spread data was extracted from the Refinitiv Eikon/LSEG Workspace Eikon through a paid subscription. The source code is available on:

<https://github.com/Aikolee0050/ImperialStatsMSC>

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A. Supplementary Materials

The optimization tradeoff in HMM regime switch:

When a hard constraint is used to maintain economic separation of the regimes, the optimizer treats the boundary as an absolute mathematical hardstop. With a soft penalty in the NLL term, the optimizer is technically allowed to cross the boundary, but pays a penalty score. To avoid that penalty, the optimizer's gradient gets distorted near the wall, as the penalty creates an artificial upward slope right before the threshold and forces the optimizer to settle at a sub-optimal parameter value just to avoid the penalty.

State Degeneracy: In a 3 regime HMM, the 3 regimes compete during the EM loop. Under hard constraints, the states are cleanly separated into boxes and cannot overlap into another regime's space. Under soft penalties, especially during windows where market data is ambiguous, the optimizer tries to allow the regimes to merge together slightly to find a local optimum and the optimizer frequently gets stuck in these suboptimal local traps vs finding a clean global fit found under hard bounds. While hard constrained optimization yields better empirical performance in terms of superior RMSE, better directional accuracy and better stability plots, using hard constraints violate asymptotic theory on paper.

Structural asymmetry in regime logic:

Oil time spreads are structurally asymmetric and do not behave the same way on the upside (extreme backwardation) vs downside (extreme contango). The downside (contango) is bounded, where a negative spread indicates oversupply. This is bounded by the 'cost of carry', which is the cost of storing oil. Once the spread covers storage and insurance, arbitrage occurs and this creates a natural floor. The upside (backwardation) is unbounded, where a positive spread represents an immediate physical shortage. Hence the 'convenience yield' can spike infinitely due to physical shortages. As the mechanics of a downward spike vs an upward spike are completely different, they cannot share the same regime or same set of parameters. As deep contango is rare and has a natural bound (storage economics), one same regime is used to handle both mild and deeper oversupply, whereas extreme backwardation is in a regime of its own.

Why do regime probabilities 'lag' actual price trends and spikes?

During major regime shifts, e.g. covid in 2020, the actual price spikes first and it takes a while for the red crisis probability to fully feed in. This is due to the Markov 'stickiness' (transition inertia). In transition matrix P , the diagonal values are very high and as such, when the market is currently in backwardation, the model's prior beliefs is that there is a very high chance of staying in the same regime in the next period. When the price spikes in the next period, the model takes in that information but it takes multiple periods of spikes before it is enough to shift the posterior probability to a new (crisis) regime (as it is also weighted by the prior). It requires sustained evidence to overpower the inertia and trigger a regime flip

Secondly, the Student-t distribution with fatter tails to model the variance affects regime shifts as well. Under a gaussian distribution, a sudden 3.0 spike is viewed as statistically impossible and would force the model to switch to crisis regime instantaneously. But as the student-t has fatter tails, in the backwardation regime, it acknowledges that the 3 jump is extreme but still mathematically possible that it is just a random 1 day tail event within the same regime. Hence

the fat tails makes the model more conservative and waits longer to see if high volatility persists before switching regimes.

Thirdly, filtering vs smoothing. The forward filter deduces the hidden state at time t using only data up to time t and nothing beyond that. Hence the model doesn't have the benefit of hindsight to know if the sudden drop is a one day anomaly or the start of a 6 month contango cycle. Alternatively a forward-backward Baum-Welch smoother uses smoothing which also looks at future data $t+1, t+2, \dots$ to retroactively correct the probabilities at time t . But this is not being adopted as in live environment, one does not have access to forward data.

Fourthly, in the `forecast_hmm_ou` function, smoothing was put in place with the inertia factor, if set to 0.8, it means the model is carrying over 80 percent of yesterday's regime probability to today.

So to reduce the lag, either the diagonals of the prior can be adjusted to make the model less sticky, OR the inertia parameter can be adjusted lower. However, the tradeoff is that reducing the lag makes the model hypersensitive and flips regimes much more easily and result in whipsawing, and false signals.

Unconstrained Optimization vs Constrained (with penalty):

In unconstrained op, the solver has complete freedom to roam the entire parameter space and finds the absolute mathematical solution that minimizes the NLL for that window. If an extreme outlier occurs, the solver is free to push μ to a high value to capture it. This achieves the best possible in-sample fit. In constrained op, it is restricting where the solver can explore. As the solver is forced to stop short of the ideal mathematical peak, the resulting NLL is always worse off. However in empirical settings, optimizing purely for best historical NLL is overfitting. A completely free optimizer will distort itself to fit every bit of noise (leading to wild parameter spikes) and memorizes the past. However, it suffers out of sample.

How regime classification is done:

Regime classification in the HMM model is handled dynamically by a Bayesian filter (composed of the `forward_filter` and `filter_update` functions), where it calculates a continuous probability distribution across all K regimes at every time step. This continuous score is represented by the vector α_t and are stored in the results as `prob_contango`, `prob_normal.backwardation` and `prob_crisis`, where all values sum to 1.0. To update this continuous score, it works in 2 sequential steps.

1. The prediction step (transition prior) - before seeing today's spread value, the model looks at yesterday's regime probabilities (α_{t-1}) and projects them forward using the transition matrix (P):

$$\text{Predicted State} = \alpha_{t-1} \times P$$

This accounts for the structural stickiness of the market

2. The update step (emission likelihood) - once today's spread values arrive, the model evaluates how surprising that price move is under each regime's OU parameters, using a student-t pdf.

Bayes rules then combines the prediction and observation by multiplying the predicted state probabilities by the observed likelihood and normalizes so they sum to 1.0

From continuous-time OU process to discrete intervals (Δt):

Vasicek (1977) formally introduces the continuous OU process to financial economics (Vasicek model) and explicitly outlines the solution using an integrating factor and derives the exact conditional mean and variance for discrete time steps Schwartz (1997) commodity specific application; adapts the continuous time mean-reverting process into discrete econometric estimation structures Gillespie (1996) a pure mathematical and stochastic-calculus breakdown. Provides the exact probability transition structure, showing how the conditional mean and variance (via Ito isometry drop out of the stochastic integral without approximation error) Kloeden and Platen (1992) chapter 4 and 9 covers the analytical solution of linear SDEs and why the Euler Maruyama approximation introduces bias and why the exact integrating-factor solution must be used for linear mean-reverting processes.

Solve the SDE analytically (exists for OU) over a finite time step (instead of using a crude Euler approximation)

$$dX_t = \kappa(\mu - X_t)dt + \sigma dW_t$$

Apply the integrating factor. To solve the linear SDE from time t to $t + \Delta t$, multiply the entire equation by the integrating factor $e^{\kappa t}$:

$$e^{\kappa t} dX_t + \kappa e^{\kappa t} X_t dt = \kappa e^{\kappa t} \mu dt + \sigma e^{\kappa t} dW_t$$

LHS of the equation is the exact derivative of the product $e^{\kappa t} X_t$:

$$d(e^{\kappa t} X_t) = \kappa e^{\kappa t} \mu dt + \sigma e^{\kappa t} dW_t$$

Then integrate over the interval $[t, t + \Delta t]$, from start time to future time.

$$\int_t^{t+\Delta t} d(e^{\kappa s} X_s) = \int_t^{t+\Delta t} \kappa e^{\kappa s} \mu ds + \int_t^{t+\Delta t} \sigma e^{\kappa s} dW_s$$

Evaluating the integrals,

$$e^{\kappa(t+\Delta t)} X_{t+\Delta t} - e^{\kappa t} X_t = \mu (e^{\kappa(t+\Delta t)} - e^{\kappa t}) + \sigma \int_t^{t+\Delta t} e^{\kappa s} dW_s$$

Then solve for $X_{t+\Delta t}$, by multiplying the entire equation by $e^{-\kappa(t+\Delta t)}$ to isolate $X_{t+\Delta t}$ on the LHS:

$$X_{t+\Delta t} = X_t e^{-\kappa \Delta t} + \mu(1 - e^{-\kappa \Delta t}) + \sigma \int_t^{t+\Delta t} e^{-\kappa(t+\Delta t-s)} dW_s$$

which yields the exact discrete evolution equation for $X_{t+\Delta t}$.

Conditional mean: Noting that the expectation of the stochastic integral term is 0:

$$\text{mean} = X_t e^{-\kappa \Delta t} + \mu(1 - e^{-\kappa \Delta t})$$

Conditional variance:

The var is derived by finding the var of the stochastic integral using Ito isometry:

$$\text{Var}(X_{t+\Delta t} | X_t) = \sigma^2 \int_t^{t+\Delta t} e^{-2\kappa(t+\Delta t-s)} ds$$

Evaluating the integral yields the exact discrete variance formula used in the M step:

$$\text{var} = \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa\Delta t})$$

Suitability of using Augmented Dickey-Fuller (ADF) test:

The standard ADF test assumes that if a time series is mean-reverting, it is reverting to a constant, stationary long-run mean. ADF test is also very sensitive to structural breaks and if the mean shifts midway through the 104-week window, the ADF test interprets that as non-stationary (unit root), even if the series is mean-reverting locally around its shifted anchors.

Establishing local stationarity across windows for one period forward

For each rolling window, a first-order autoregressive (AR(1)) model was fit via OLS:

$$X_t = \alpha + \phi X_{t-1} + \epsilon_t$$

where α is a constant drift term, ϕ represents the autoregressive persistence parameter, and ϵ_t is the white-noise error term.

The persistent parameter ϕ indicates how much of the previous period's deviation from equilibrium survives the next period.

As the empirical estimates mostly satisfy the condition

$$0 < \hat{\phi}_t < 1$$

consistently across the rolling windows, it supports local stationarity and mean reverting behavior, implying that the dynamics of oil spreads follow a bounded, mean-reverting process.

To translate the persistence parameter ϕ into an economically intuitive metric, the half-life (HL_t), is derived from the OU process, which measures the expected duration (in weeks) required for a shock to revert midway back to its equilibrium. It is derived analytically as

$$HL_t = \frac{\ln(0.5)}{\ln(\hat{\phi}_t)} = \frac{-\ln(2)}{\ln(\hat{\phi}_t)}$$

. When $\hat{\phi}_t \rightarrow 1.0$, $HL_t \rightarrow \infty$, indicating non-stationary or structural market breakdowns.

Comparison between HMM-OU model vs student-t OU model

This section investigates how the single-state student t OU model and regime-switching HMM OU processes extreme crisis events in history. The March 2020 Covid event is used to illustrate the difference in how the two models handle crisis events

During the March 2020 period, Brent time spreads collapsed into deep contango (-\$3.50)

In the regime switching HMM-OU model, the trailing 80-week lookback window before the crisis was quiet and range-bound in 2018-2019, hence the HMM filter assigned high probability to State 0 (low volatility and tight equilibrium center), with narrow confidence bands. When the Brent spread collapsed to (-\$3.50), the forward filter computes the likelihood of this observation across all three states :

$$\mathcal{N}(X_{\text{actual}} \mid \mu_k, \sigma_k)$$

For state 0, a drop to -\$3.50 is statistically impossible given its narrow volatility ($\sigma = 0.40$), hence the probability of being in state 0 falls to 0. State 2 is assigned much higher probability as $\sigma \approx 6.80$ (Figure 5). The confidence intervals widen massively

In contrast, the student-T OU model does not rely on multiple states but a single state with a heavy-tailed error distribution to absorb the shocks. The rolling lookback window also allows for dynamic adjustment of the parameter shapes $(\theta, \mu, \sigma, \nu)$. A high ν implies that the error distribution behaves almost like a standard Gaussian curve, under calm market conditions with low volatility. When the shock hits in March 2020, as there is no alternative regimes to route this extreme data point to, the single-state OU is forced to absorb this anomaly by driving the degrees-of-freedom parameter ν down to ($\nu \approx 2.5$). By dropping ν to 2.5, the model enters the heavy-tailed region, expanding the forecast cone width to absorb the widening volatility and the model registers it as a breakdown in mean reversion during this period and reverts to a random walk until the rolling 80-week variance stabilizes again.

Algorithm 2 Weighted MAP Parameter Estimation

```

1: function FIT_OU_WEIGHTED( $X, \Delta t, w, \text{regime\_idx}$ )
2:   define Neg-Log-Posterior( $\theta = [\kappa, \mu, \sigma]$ ):
3:     if  $\kappa \leq 0$  or  $\sigma \leq 0$  then return  $10^{10}$ 
4:   end if
5:   Compute exact discrete OU mean and Student- $t$  scaled variance over  $\Delta t$ 
6:    $\ell_{\text{base}} \leftarrow -\sum w[: -1] \cdot \log t_{\text{pdf}}(X[1 :] \mid \text{mean, scale, } \nu)$ 
7:    $\mathcal{L}_{\text{prior}} \leftarrow 0$ 
8:   if  $\text{regime\_idx} == 0$  then ▷ Contango Regime
9:      $\mathcal{L}_{\text{prior}} += \text{Half-Normal}(\mu, \text{contango\_threshold}) + \text{Inv-Gamma}(\sigma)$ 
10:  else if  $\text{regime\_idx} == 1$  then ▷ Normal Regime
11:     $\mathcal{L}_{\text{prior}} += \text{Half-Normal}(\mu, [0.01, 3.0]) + \text{Half-Normal}(\sigma, 6.0)$ 
12:  else if  $\text{regime\_idx} == 2$  then ▷ Crisis Regime
13:     $\mathcal{L}_{\text{prior}} += \text{Half-Normal}(\mu, \text{crisis\_threshold})$ 
14:  end if
15:  return  $\ell_{\text{base}} - \mathcal{L}_{\text{prior}}$ 
16:  return Minimize-L-BFGS-B(Neg-Log-Posterior,  $\theta_{\text{init}}$ , bounds)
17: end function

18: function FIT_SINGLE_OU( $X, \Delta t$ )
19:    $\mathbf{w} \leftarrow \mathbf{1}_{\text{length}(X)}$ 
20:   return FIT_OU_WEIGHTED( $X, \Delta t, \mathbf{w}, \text{regime\_idx} = \text{None}$ )
21: end function

```

“It was the best of times, it was the worst of times, it was the age of wisdom, it was the age of foolishness, it was the epoch of belief, it was the epoch of incredulity, it was the season of Light,

Algorithm 3 HMM-OU State Estimation & Expectation-Maximization

```

1: function FORWARD_FILTER( $X, \{\theta_k\}, P, \Delta t$ )
2:    $\alpha \leftarrow \mathbf{0}_{T \times K}$ ,  $\alpha[0] \leftarrow \frac{1}{K}$ ,  $\mathcal{L}_{\text{total}} \leftarrow 0$ 
3:   for  $t = 0$  to  $T - 2$  do
4:     for  $i = 0$  to  $K - 1$  do
5:        $\text{obs\_probs}[i] \leftarrow \text{Student-}t\text{-PDF}(X[t + 1] \mid X[t], \theta_i, \Delta t)$ 
6:     end for
7:      $\text{pred} \leftarrow \alpha[t]P$ ,  $\text{raw} \leftarrow \text{pred} \odot \text{obs\_probs}$ 
8:      $s \leftarrow \sum \text{raw}$ ,  $\alpha[t + 1] \leftarrow \text{raw}/s$ ,  $\mathcal{L}_{\text{total}} += \log(s)$ 
9:   end for
10:  return  $\alpha, \mathcal{L}_{\text{total}}$ 
11: end function

12: function FIT_HMM_OU( $X, \Delta t, N_{\text{iter}}$ )
13:  Initialize  $\{\theta_k\}$  for  $K = 3$  via percentiles,  $P \leftarrow P_{\text{prior}}$ 
14:  for  $\text{iter} = 1$  to  $N_{\text{iter}}$  do
15:     $\alpha, \mathcal{L} \leftarrow \text{FORWARD\_FILTER}(X, \{\theta_k\}, P, \Delta t)$  ▷ E-Step
16:    for  $k = 0$  to  $K - 1$  do
17:       $\theta_k \leftarrow \text{FIT\_OU\_WEIGHTED}(X, \Delta t, \alpha[:, k], k)$  ▷ M-Step Parameters
18:    end for
19:    Compute joint transition counts  $A_{\text{num}}$ 
20:    for  $i = 0$  to  $K - 1$  do
21:       $P[i] \leftarrow (1 - \eta) \left( \frac{A_{\text{num}}[i]}{\sum_j A_{\text{num}}[i, j]} \right) + \eta P_{\text{prior}}[i]$  ▷ Inertia Shrinkage
22:    end for
23:    Normalize rows of  $P$  and sort  $\{\theta_k\}, P, \alpha$  by  $\mu_k$  ascending
24:  end for
25:  return  $\{\theta_k\}, P, \alpha$ 
26: end function

```

Algorithm 4 Single and Mixture HMM-OU Forecasting

```

1: function FORECAST_SINGLE_OU( $x_{\text{today}}, \theta, \Delta t, h$ )
2:   Extract  $(\kappa, \mu, \sigma)$  from  $\theta$ 
3:    $\text{mean} \leftarrow \mu + (x_{\text{today}} - \mu)e^{-\kappa h}$ 
4:    $\text{var} \leftarrow \left( \frac{\sigma^2}{2\kappa} (1 - e^{-2\kappa h}) \right) \cdot \frac{\nu}{\nu - 2}$ 
5:   return  $\text{mean}, \sqrt{\max(\text{var}, 10^{-8})}$ 
6: end function

7: function FORECAST_HMM_OU( $x_{\text{today}}, \{\theta_k\}, \alpha_t, P, \Delta t, h, h_{\text{steps}}, \pi_{\text{prev}}$ )
8:    $\pi_{\text{raw}} \leftarrow \alpha_t \cdot P^{h_{\text{steps}}}$ 
9:    $\mathbf{w} \leftarrow (1 - \eta)\pi_{\text{raw}} + \eta\pi_{\text{prev}}$  ▷ Exponential Smoothing
10:  Zero out weights  $< 0.01$  (Dormant State Pruning) and re-normalize  $\mathbf{w}$ 
11:  for  $k = 0$  to  $K - 1$  do
12:    Compute  $h$ -step mean  $\text{mean}_k$  and variance  $\text{var}_k$  for regime  $k$ 
13:  end for
14:   $\mu_{\text{mix}} \leftarrow \sum w_k \text{mean}_k$ 
15:   $\sigma_{\text{mix}}^2 \leftarrow \sum w_k (\text{var}_k + \text{mean}_k^2) - \mu_{\text{mix}}^2$ 
16:  return  $\mu_{\text{mix}}, \sqrt{\max(\sigma_{\text{mix}}^2, 10^{-8})}, \mathbf{w}$ 
17: end function

```

Table 9.: Piecewise Quadratic Boundary Parameters and Scale Constants (a)

Regime	Parameter (z)	Boundary (b)	Direction	Scale Parameter (a)
Contango (0)	μ_0	Contango Threshold (c_0)	Upper	0.050
Contango (0)	σ_0	Contango Vol Cap (v_0)	Upper	0.100
Normal (1)	μ_1	0.01	Lower	0.007
Normal (1)	μ_1	3.00	Upper	0.007
Normal (1)	σ_1	6.00	Upper	0.022
Crisis (2)	μ_2	Crisis Threshold (Q_{95})	Lower	0.007
Crisis (2)	μ_2	1.00	Lower	0.022

it was the season of Darkness, it was the spring of hope, it was the winter of despair, we had everything before us, we had nothing before us, we were all going direct to Heaven, we were all going direct the other way - in short, the period was so far like the present period, that some of its noisiest authorities insisted on its being received, for good or for evil, in the superlative degree of comparison only.” (?)

B. Definitions, theorems and examples

Solving the OU SDE:

Starting with the continuous-time OU SDE

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t$$

Integrate from time t to $t + h$, use u as dummy time variable for the integration steps:

$$dx_u = \theta(\mu - x_u)du + \sigma dW_u$$

Rearrange the terms to group the x_u variables on the left:

$$dx_u + \theta x_u du = \theta \mu du + \sigma dW_u$$

Multiply the entire equation by the integrating factor $e^{\theta u}$:

$$e^{\theta u} dx_u + \theta e^{\theta u} x_u du = \theta \mu e^{\theta u} du + \sigma e^{\theta u} dW_u$$

Using Itô's Lemma (the stochastic product rule), the left side is the differential of the product $x_u e^{\theta u}$. As $e^{\theta u}$ is deterministic, no extra Itô terms are generated:

$$d(x_u e^{\theta u}) = e^{\theta u} dx_u + \theta x_u e^{\theta u} du$$

Substitute back into the equation:

$$d(x_u e^{\theta u}) = \theta \mu e^{\theta u} du + \sigma e^{\theta u} dW_u$$

Integrate both sides over the future horizon, starting from current time t and ending at future time $t + h$.

The Left Side (State Variable):

$$\int_t^{t+h} d(x_u e^{\theta u}) = x_{t+h} e^{\theta(t+h)} - x_t e^{\theta t}$$

The Right Side (Drift):

$$\int_t^{t+h} \theta \mu e^{\theta u} du = \theta \mu \left[\frac{1}{\theta} e^{\theta u} \right]_t^{t+h} = \mu (e^{\theta(t+h)} - e^{\theta t})$$

The Right Side (Diffusion):

This remains a stochastic Itô integral with respect to Brownian motion:

$$\int_t^{t+h} \sigma e^{\theta u} dW_u$$

Putting the integrated parts together:

$$x_{t+h} e^{\theta(t+h)} - x_t e^{\theta t} = \mu (e^{\theta(t+h)} - e^{\theta t}) + \int_t^{t+h} \sigma e^{\theta u} dW_u$$

Solving for the Future State (x_{t+h})

To isolate x_{t+h} , multiply the entire equation by $e^{-\theta(t+h)}$:

$$x_{t+h} - x_t e^{-\theta h} = \mu (1 - e^{-\theta h}) + \int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u$$

Move the x_t term to the right side and rearrange the deterministic terms:

$$x_{t+h} = x_t e^{-\theta h} + \mu - \mu e^{-\theta h} + \int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u$$

Factor out the $e^{-\theta h}$:

$$x_{t+h} = \mu + (x_t - \mu) e^{-\theta h} + \int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u$$

Deriving the mean (conditional expectation) for the forecast:

To find the expected future value given today's value x_t , take the expectation $\mathbb{E}[\cdot|x_t]$ of both sides. The Itô integral of a deterministic function against Brownian motion is a martingale, so its conditional expectation is exactly zero.

$$\mathbb{E}[x_{t+h}|x_t] = \mu + (x_t - \mu)e^{-\theta h} + 0$$

$$\mathbb{E}[x_{t+h}|x_t] = \mu + (x_t - \mu)e^{-\theta h}$$

Conditional variance (uncertainty)

To find the variance of the forecast, the deterministic terms (μ and x_t) drop out, leaving only the stochastic integral:

$$\text{Var}(x_{t+h}|x_t) = \text{Var} \left(\int_t^{t+h} \sigma e^{-\theta(t+h-u)} dW_u \right)$$

Using Itô's Isometry ($\mathbb{E}[(\int f(u)dW_u)^2] = \int f(u)^2 du$), turn the stochastic integral into a standard Riemann integral:

$$\text{Var}(x_{t+h}|x_t) = \int_t^{t+h} \sigma^2 e^{-2\theta(t+h-u)} du$$

Evaluating the integral yields the exact formula used in the backtest to map the expanding variance cone over horizon h :

$$\text{Var}(x_{t+h}|x_t) = \frac{\sigma^2}{2\theta} (1 - e^{-2\theta h})$$